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Review

High-order harmonic generation and other intense optical field–matter interactions: review of recent experimental and theoretical advances

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Abstract

An overview of recent experimental and theoretical developments in high-order harmonic generation (HHG), above threshold ionization (ATI), and other phenomena originating in intense optical field-matter interactions, is attempted here. A recurring theme in this review is the common bond between HHG, ATI, laser-assisted Auger decay, and the laser-assisted photoelectric effect, which is attributable solely to the photon field–electron interaction. It is argued that the dominant mechanism for photon production in HHG is not the recombination of an ion with a fast electron but rather is the result of coupling between the optical field and an electron that generates photons having energies that are an integral multiple of the pump photon energy. Characterized and steadily developed over the past decade, HHG and its attendant technology have matured to the extent that it now offers a versatile, table top source of coherent radiation in the vacuum ultraviolet to X-ray spectral regions with the capability of producing pulse widths as short as several hundred attoseconds. Combined with its comparative simplicity, the availability of average powers exceeding 1 mW from a single (plateau) harmonic at pulse repetition frequencies >1 kHz make HHG an attractive alternative to other nonlinear sources requiring tunable dye lasers and metal vapors. Emphasis is placed on the optical and spatial characteristics of the harmonic radiation and their potential impact on quantum electronics.
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Contents

1. Introductory comments: an emerging tool for quantum electronics in the ultraviolet to X-ray regions . . . 198
2. Historical background . . . 200
2.1. Ultraviolet and vacuum ultraviolet coherent sources driven by nonlinear optical processes in metal vapors . . . 200
2.2. Early experiments in harmonic generation by irradiation of gases, plasmas, and solid targets . . . 201
3. Review of recent experimental results . . . 203
3.1. High-order harmonic generation: gaseous and solid targets . . . 203
3.1.1. Spectra: photon energies beyond 500 eV . . . 204
3.1.2. Conversion efficiency of HHG . . . 207
3.1.3. Temporal, spatial, spectral, and polarization characteristics of harmonic radiation . . . 211
3.1.4. Attosecond pulse generation . . . 221
3.1.5. Experimental considerations . . . 222
3.1.6. Summary . . . 226
3.2. Above threshold ionization (ATI), laser-assisted Auger decay (LAAD), and the laser-assisted photoelectric effect (LAPE) . . . 227
4. Theory . . . 231
4.1. Semiclassical picture: two step process . . . 231
4.2. Optical transitions between stationary states of the photon field-electron system: quantized optical field view of HHG . . . 234
4.2.1. General Comments . . . 234
4.2.2. Quantum Volkov state theory of harmonic generation 236
5. Summary and conclusions . . . 241
Acknowledgements . . . 242
References . . . 242

1. Introductory comments: an emerging tool for quantum electronics in the ultraviolet to X-ray regions

Developing primary coherent sources in the ultraviolet (UV) to X-ray spectral regions has proven to be a daunting task for several reasons, one of which is the well-known λ^{-4} scaling of pump power with wavelength, λ (Ref. [1]). This statement is particularly true for wavelengths below 200 nm where few powerful and efficient sources exist (see, for example, Refs. [2] and [3]). With the exception of the ArF (193 nm) and F_2 (158 nm) lasers, adoption of the atomic or molecular oscillators presently available (such as the Xe^{2+} transition at 108.9 nm (Refs. [4–6]) and the Ar laser at $\sim$47 nm (Ref. [7])) is neither feasible nor attractive for all but a limited number of laboratories worldwide, and none have thus far been engineered to the

point of being commercially available. It is only natural that persistent barriers to developing primary sources would inevitably divert attention into pursuing nonlinear processes for generating short wavelength, coherent radiation. Although it was recognized soon after the advent of the laser that resonant enhancements of the third-order nonlinear susceptibility $\chi^{(3)}$ could be exploited to efficiently produce coherent UV and vacuum ultraviolet (VUV) radiation, these early efforts provided only a modest upshift in the photon energy ($3 \times \hbar\omega_{\text{pump}}$) because of the order of the nonlinearity.

The developments described and pondered in this review were precipitated largely by the introduction of high peak power, sub-picosecond lasers. Foremost among these is the Ti-doped sapphire (Ti:Al_2O_3) laser [8] which is capable of producing, from a moderate, commercially available system, sub-100 fs pulses with mJ pulse energies at pulse repetition frequencies (PRFs) $\geqslant 1$ kHz. It is not an overstatement to conclude that this laser has revolutionized high field nonlinear optics because the peak optical field intensities that can now be generated routinely ($10^{15}-10^{18}$ W cm^{-2}) provide access to optical processes having a degree of nonlinearity that would have been unimaginable just 20 years ago.

Few phenomena have captured the attention and energies of optical physicists over the past decade as has high-order harmonic generation (HHG), the term given to the production of harmonics of the driving field when an intense optical field (typically $I > 10^{14}$ W cm^{-2}) interacts with a gas, plasma, or solid target. Harmonic orders beyond 300 and wavelengths well into the soft X-ray region have now been generated in several laboratories and, with careful attention given to phase matching and focusing geometry, average powers of milliwatts *from a single harmonic* can now be realized for laser PRFs of 1 kHz. Rapid progress in the optimization of the HHG process (energy/harmonic/pulse, and spatial and optical characteristics of the output radiation) over the past several years reinforces the conviction that HHG has matured to the point of being a viable quantum electronic tool for physical, chemical, medical, and materials research in the VUV to X-ray regions. One thesis of this review is that the impact of HHG sources on research and optical engineering at short wavelengths is likely to rival that of the N_2 laser (337 nm) in an earlier era. For more than a generation, the simplicity of the N_2 laser placed an inexpensive and reliable source of coherent UV within the reach of researchers in the life and physical sciences. While it is certainly true that the complexity and cost of Ti:Al_2O_3 systems generally exceed those for an N_2 laser by at least an order of magnitude, both are dwarfed by the few alternative sources (e.g., synchrotrons) that offer wavelength agility at high photon energies ($\hbar\omega > 10$ eV).

Consequently, HHG is now a reliable, reasonably compact (table top) source of coherent radiation and is the centerpiece of the discussion to follow. Other reviews of high field nonlinear optical phenomena, such as the excellent synopsis of few-cycle laser fields and their impact on nonlinear optics and high field physics by Brabec and Krausz [9], have appeared in the literature but emphasis here is placed on the utility of HHG as a short wavelength coherent source. Section 2 offers a perspective on recent developments by reviewing early experimental efforts to generate coherent UV and VUV radiation by resonantly enhanced nonlinear processes. Following the

startling announcement in 1986 by C. K. Rhodes' research group of the observation of the 17th harmonic of 248 nm, several groups worldwide vied to generate still higher harmonics and these efforts are also summarized briefly in Section 2. More recent results, almost all of which resulted from excitation of gaseous and solid targets with the Ti:Al_2O_3 laser, are presented in Section 3 with emphasis given to the control over the dominant harmonic and the optical and spatial characteristics of the radiation that is afforded by HHG. Section 3 includes a description of other high optical field phenomena, such as above threshold ionization (ATI) and the laser-assisted photoelectric effect (LAPE), and their clear linkage to HHG. A detailed understanding of the physical mechanisms underlying HHG is not only of intrinsic value but is indispensable for identifying avenues by which further inroads into improving HHG sources might be realized. The nonlinearities driven by the extraordinary laser intensities characteristic of HHG are clearly beyond the reach of traditional perturbative treatments of nonlinear optical phenomena, and Section 4 concludes by discussing two contrasting theoretical views of HHG. It is hoped that this review, although not exhaustive, conveys to the newcomer to the field a sense of the excitement regarding recent advances and the pace of progress.

2. Historical background

2.1. Ultraviolet and vacuum ultraviolet coherent sources driven by nonlinear optical processes in metal vapors

Efforts to generate short wavelength ($\lambda \lesssim 350$ nm) coherent radiation by nonlinear frequency upconversion from a fixed wavelength or tunable primary source have a long history, dating to before 1970. In 1967, New and Ward [10] reported the observation of the third harmonic ($\lambda = 231.4$ nm) of the ruby laser, generated in any of a variety of gases, and noted the extinction of the harmonic when the pump pulse radiation was circularly polarized. Four years later, Harris and colleagues at Stanford showed [11, 12] that the efficiency of third harmonic and sum frequency generation in alkali metal vapors is enhanced by phase matching through the addition of a rare gas. For at least the next decade, the attention of the nonlinear optics community focused on generating coherent UV and VUV radiation in metal vapors, owing to the large atomic transition moments that are available with, for example, the alkalis, and Groups IIA and IIB elements.

Broad tunability in the VUV, however, was not available until 1974 when Hodgson, Sorokin, and Wynne [13] reported that coherent VUV radiation, tunable over a range exceeding 0.4 eV in breadth in the 180 nm region, could be produced by a four-wave mixing process ($2\omega_1 + \omega_2$) in Sr vapor. The seminal contribution of these experiments was the realization that the overall transition matrix element for the $\chi^{(3)}$ process is magnified by the presence of a real state in the vicinity of $2\hbar\omega_1$. Over the next several years, the experiments of Ref. [13] were extended to other metal vapors and, by 1985, continuous tunability in the VUV from 104.5 to $\sim$210 nm was available with the vapors of Hg, Mg, and Zn, as well as the rare gases Kr and Xe.

Highlights of this period include the demonstration by McKee, Stoicheff, and Wallace [14] of tunability around the Lyman-α transition (121.6 nm) in Mg vapor, generating coherent VUV to the LiF cutoff at 104.5 nm (Ref. [15]; for further details, see Ref. [16]), and the attainment by Tomkins and Mahon [17] of pump-to-output conversion efficiencies in Hg vapor as high as 1% in the 121.9–125.5 nm region. With a collimated beam geometry and optical field-nonlinear medium interaction lengths >1 m, Muller et al. [18] subsequently realized an energy conversion efficiency of 5% in a four wave mixing scheme in which three pump wavelengths produced coherent radiation at 130 nm. These results, including output intensities $>2\,\mathrm{MW\,cm^{-2}}$ and a continuous tunability of $2.5\,\mathrm{cm^{-1}}$, represent the highest efficiency achieved to date for the deep-VUV. It must be emphasized that, prior to these advances, few primary sources existed in the UV and VUV (H_2, CO, Xe_2, Kr_2, N_2, rare gas-halides, etc.) and their tunability was severely limited. Consequently, nonlinear frequency upconversion processes provided in the VUV what primary sources could not — coverage of the entire spectral region. Resonantly enhanced multiwave mixing and harmonic generation in the metal vapors and rare gases have continued to be the workhorse of spectroscopy and photochemistry in the VUV and portions of the UV, despite the fact that, since the late 1980s, conversion efficiencies have remained essentially unchanged [19].

2.2. *Early experiments in harmonic generation by irradiation of gases, plasmas, and solid targets*

Initiatives in several national laboratories in the 1970s to demonstrate inertial confinement fusion fueled the construction of high peak power, sub-nanosecond CO_2 and Nd:glass laser systems designed to explore laser-solid target interactions. Although the second harmonic and the fractional harmonic 3/2 ω_0 were observed routinely in such experiments, Burnett et al. [20] were the first to report, in 1977, the detection of harmonics higher than the third. In experiments with Al targets and CO_2 laser intensities $>10^{14}\,\mathrm{W\,cm^{-2}}$, they observed harmonics extending from the third up to $q=11$, where q denotes the harmonic order. Concurrently, McLean and coworkers [21] at the Naval Research Laboratory generated the third, fourth, and fifth harmonics of an Nd:glass laser by focusing 6–10 J, 75 ps pulses onto polystyrene targets. In subsequent experiments with CO_2 laser intensities exceeding $5\times10^{14}\,\mathrm{W\,cm^{-2}}$, and plastic and metal-coated glass microballoon targets, Carman, Forslund, and Kindel [22] observed harmonics (more than 10 of which lie in the visible) at least as high as the 29th and coined the term "cutoff" to refer to the rapid decline in intensity for harmonics above a critical value of q. In describing both experimental and calculated harmonic spectra, they state [22] "…we see a flattening of the harmonic emission above $2\omega_0$ and a sharp cutoff at a high harmonic number… The striking feature is the nearly constant efficiency of generation of the higher harmonics until cutoff." At still higher CO_2 laser intensities ($\lesssim3.5\times10^{15}\,\mathrm{W\,cm^{-2}}$), Carman et al. [23] acquired spectra showing the 46th harmonic (~230 nm) and half-integer harmonics up to 87/2. Between the 16th and 42nd harmonics, in a spectral region that has come to be known as the plateau, the harmonic intensity was approximately constant but fell quickly for $\lambda<252$ nm.

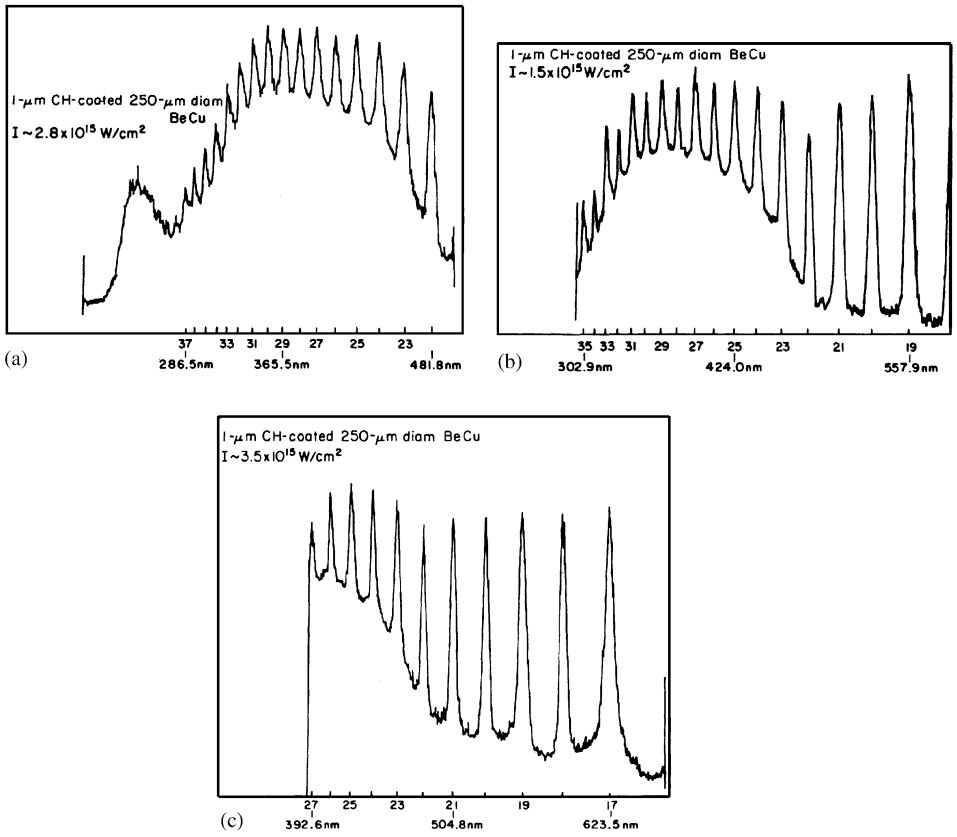


Fig. 1. Ultraviolet and visible spectra (derived from microdensitometer traces of photographic negatives) of the harmonics produced by CO_2 laser irradiation of polyethylene-overcoated BeCu wires (reprinted from Ref. [23] by permission).

Spectra representative of those reported in Ref. 23 for polyethylene-coated BeCu wire targets are presented in Fig. 1. Bezzerides et al. [24] proposed that the interaction of resonant absorption with a dense plasma having a large gradient in number density generates a restoring force that provides the nonlinearity necessary to account for these results. Several theoretical viewpoints that have been advanced to explain these and other harmonic generation experiments are described more fully in Section IV.

During this same time period, others demonstrated the feasibility of producing harmonics in gases and vapors. With a frequency-quadrupled, 30 ps Nd:YAG laser system, Reintjes and colleagues [25, 26], (for a review of early developments, see Ref. [25]) generated the shortest wavelength coherent radiation available at that time (late 1970s) by producing, in the rare gases, the fifth and seventh harmonics of 266 nm (53 and 38 nm, respectively). Grozeva et al. [27] reported observing the ninth harmonic of Nd:glass (117.7 nm) in Na vapor and, following the introduction of the

rare gas-halide lasers, several studies [28, 29] were conducted with XeCl (308 nm) and KrF (248 nm) laser drivers producing pulses as short as 15 ps. In Ref. [29], the production of coherent radiation at 35.5 nm, the seventh harmonic of the KrF laser, in He was described. With an iodine laser, generating 750 ps pulses at 1.315 μm, Wildenauer [30] extended to 11 the maximum harmonic order observed in the rare gases.

Events took an unexpected turn when, in 1987, McPherson et al. [31] reported generating the *17th harmonic of the KrF laser (14.6 nm)* by irradiating Ne gas with peak laser intensities of 10^{15}–10^{16} W cm^{-2}. These results more than doubled the highest photon energies that had been produced by harmonic generation. The authors noted the presence of a plateau above the 11th harmonic, thereby confirming the general conclusions of Ref. [23], published several years earlier, regarding CO_2 laser-produced HHG spectra. The clear demonstration of the ability of HHG to produce coherent radiation at wavelengths on the threshold of the soft X-ray region provided the impetus for a number of experimental groups worldwide to pursue this promising phenomenon. Ferray and co-workers [32] and Li et al. [33] of Saclay quickly reinforced the observations of the University of Illinois group (Ref. [31]) by recording the presence of harmonics as high as 33, as well as a distinct plateau, when the medium was again Ne but the optical field wavelength was 1.06 μm.

As the 1980s came to a close, several conclusions regarding HHG were already unavoidable, the foremost of which is that for driving optical field intensities exceeding $\sim 10^{14}$ W cm^{-2}, HHG is ubiquitous, having been observed with a variety of both gaseous and solid targets. Even and odd harmonics are observed for solid targets whereas, for reasons of symmetry, only odd harmonics are generated in gases by a monochromatic driving field. Radiation at wavelengths approaching 10 nm had been produced with laser wavelengths of 1.06 μm and 248 nm and pulse widths of 36 ps and ~ 1 ps, respectively. Most striking was the extraordinarily high order of the harmonics that had been observed, underscoring the conviction in the community that nonlinear optics had entered a new era in which the perturbative view of harmonic generation was no longer valid. Every indication pointed to the feasibility of realizing experimentally still higher energy photons if the process was optimized with respect to the laser parameters, the gas target and its pressure. These expectations were confirmed emphatically (indeed, surpassed) in the next decade.

3. Review of recent experimental results

3.1. High-order harmonic generation: gaseous and solid targets

The startling progress made in the past dozen years in characterizing HHG and relentlessly driving the maximum attainable photon energy further into the X-ray region is the subject of this section. Although recent experiments involving solid (and liquid) targets will be mentioned, the discussion will be skewed decidedly towards gaseous media, because of the breadth of data available and the comparative

experimental simplicity and convenience (absence of target debris, self-healing, etc.) of gases.

3.1.1. Spectra: photon energies beyond 500 eV

The pace of developments increased dramatically in the early 1990s. In a rapid succession of experiments, groups in Japan, Europe, the US and elsewhere explored several different wavelengths and pulse widths for the driving optical field. Sarukura et al. [34] reported harmonics of the KrF laser, produced in He and Ne with 500 fs pulses, as high as 25. Shortly thereafter, the first experiments with a dye laser (616 nm, 800 fs) were described by Miyazaki and Sakai [35] who observed the 41st harmonic ($\lambda = 15$ nm) in He. That same year (1992), Crane and colleagues [36] at the Lawrence Livermore National Laboratory detected the 45th harmonic of 526 nm in experiments with He in which the laser was a 600 fs, frequency-doubled Nd:glass system.

A major advance occurred in 1993 when Macklin et al. [37] at Stanford announced the detection of harmonics beyond 100 (maximum of ≈ 109) when a 15 mJ per pulse, Ti:Al_2O_3 (sapphire) laser system ($\lambda \sim 806$ nm) provided an optical field having a peak intensity $\gtrsim 10^{15}$ W cm^{-2}. Reducing the laser pulse width to 125 fs allowed for high harmonics to be produced "…at a low degree of ionization," [37] leading to the generation of sub-10 nm wavelength coherent radiation by HHG for the first time. L'Huillier and Balcou simultaneously reported [38] producing harmonics in Ne as high as 135 from a 1 ps, 1.06 μm source but the conclusions of Ref. [37] strongly suggested that the most productive avenue for future improvements in HHG and the efficient generation of coherent radiation in the XUV and soft X-ray regions, in particular, lay in pursuing shorter (< 100 fs) laser pulse widths so as to expose an atom to an intense optical field with minimal rise time. Suppressing ionization in the active medium prior to the appearance of the peak field intensity implies not only pulses as short as possible but gaseous media with the maximum available ionization potentials (e.g., Ne and He).

Kondo and co-workers [39] affirmed these early indications by comparing the efficiency of HHG for both the Ti:sapphire and KrF laser systems. When producing 100 fs pulses, the KrF laser generated no detectable high harmonic radiation in the rare gases above the 11th harmonic, owing to severe self-phase modulation (SPM) and the associated chirp that can quickly degrade phase matching for the higher harmonics. Experiments with a 120 fs Ti:sapphire system, on the other hand, yielded HHG spectra with clear plateaus extending to the 63rd harmonic (in Ne) for a peak intensity of 3.1×10^{15} W cm^{-2}. Experiments conducted by Christov et al. [40] revealed that as many as 20 additional harmonic orders were detectable when the laser intensity was fixed but the pulse width was reduced from 100 to 25 fs. Simulations showed these results to be consistent with the premise that, with the shorter pulse durations, ionization of the atom is forestalled until higher fundamental field intensities are reached. Although work continued for several years with other lasers (such as the Nd-doped solid state systems and the rare gas-halide excimers) restricted to longer pulses, virtually all activity in the arena of HHG experiments quickly narrowed to the Ti:sapphire system, focussing on reducing its

pulse width and, eventually, on controlling carefully the phase characteristics of the optical field.

In the years immediately following the seminal experiments of Refs. [37, 38], the 10 nm milestone in the wavelength of generated harmonics was traversed with regularity. The 141st harmonic (7.5 nm) of an Nd:glass (1.053 μm) laser was produced in He by Perry and Mourou [41], and Preston et al. [42] reported harmonic radiation wavelengths down to 6.7 nm (37th harmonic of 248 nm) with a KrF laser generating focal intensities of $4 \times 10^{17}\,\mathrm{W\,cm^{-2}}$ in 380 fs pulses. However, it was the realization of the long-sought goal of producing coherent soft X-ray radiation in the "water window" (2.3–4.4 nm) that captured the attention of the HHG community. With a Ti:sapphire laser system producing 20 mJ ($I \sim 6 \times 10^{15}\,\mathrm{W\,cm^{-2}}$), ~25 fs pulses (the latter of which is ~20% of the pulse duration in Ref. [37]), Chang et al. [43] observed discrete harmonics as high as 221 and 155 in He and Ne, respectively. Corresponding to wavelengths (photon energies) of 3.6 nm (340 eV) and 5.2 nm (239 eV), respectively, such harmonic orders represent a factor of two increase over those obtained with pulses 125 fs in duration [37]. The shortest wavelength radiation observed in these 1997 experiments (2.7 nm in He) [43] is associated with unresolved emission from harmonic number 297. Prior experiments [44] by the same research group with the ~25 fs system had observed unexpectedly high harmonics in the heavier rare gases (Ar, Kr, and Xe) but the detection system complicated similar studies of harmonics generated in Ne and He above $q \sim 135$.

At about the same time, Spielmann and colleagues [45] also generated, with a sub-10 fs Ti:sapphire system, coherent continuum extending into the water window. In subsequent experiments [46, 47], the production of photon energies beyond 500 eV ($\lambda < 2.5$ nm) was observed when He was irradiated by sub-10 fs, <1 mJ pulses at an intensity of $\sim 3.5 \times 10^{15}\,\mathrm{W\,cm^{-2}}$. The data suggested that the $\lesssim$10 fs duration of the pulses is critical for obtaining photons having energies beyond 400 eV, and Schnürer et al. [46] concluded that the "…X-ray yield in the vicinity of the carbon K edge at 280 eV of approximately 10^6 photons/s in a 10% bandwidth is already sufficient to prompt consideration of practical applications for this coherent water-window emission." It was also noted in Ref. [46] that the X-ray photon yield in the water window in their experiments was comparable to that reported in Ref. [43]. Considering these developments, it is remarkable that, in the span of roughly one decade, the frontier of HHG progressed from the 17th harmonic of the KrF laser (14.6 nm), produced by a low PRF system, to significant photon yields in the X-ray water window with table-top Ti:sapphire systems operating at a PRF of 1 kHz. Although not the subject of the present review, sustained progress in Ti:sapphire laser technology (chirped dielectric mirrors, chirped pulse amplification and Kerr lens mode-locking) is largely responsible for these advances.

Harmonic spectra that are representative of those observed for excitation of the rare gases with laser pulses of <50 fs duration are shown in Fig. 2. Reprinted from Poletto et al. [48], these spectra were recorded when a He gas jet was irradiated with either 30 fs, 390 μJ or sub-10 fs, 270 μJ pulses from a Ti:sapphire ($\lambda \sim 795$ nm) system. The peak laser intensities were $3.5 \times 10^{14}\,\mathrm{W\,cm^{-2}}$ and $9 \times 10^{14}\,\mathrm{W\,cm^{-2}}$ for the 30 fs and sub-10 fs pulse duration experiments, respectively. Notice that, in contrast with

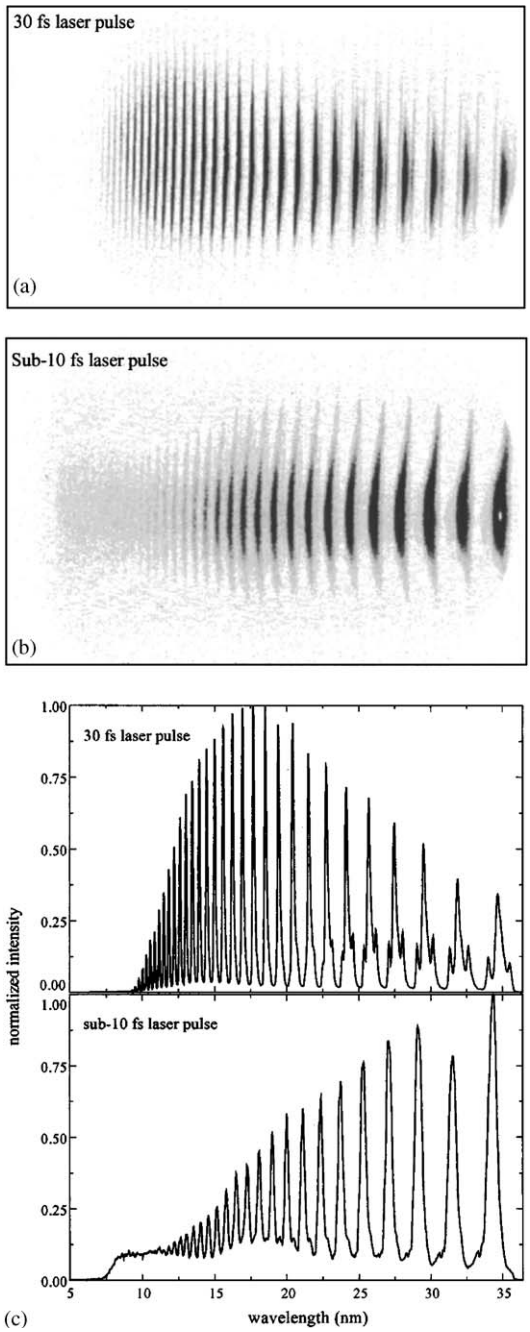


Fig. 2. HHG spectra produced in He: (a), (b) CCD images of harmonic spectra in the 6–37 nm region generated by 30 fs, 390 μJ and sub-10 fs, 270 μJ pulses, respectively. The lower portion (part (c)) shows intensity profiles (line-outs) of the false-color images of parts (a) and (b) (reprinted by permission from Ref. [48].)

harmonic spectra observed from solid targets (Fig. 1), only the odd harmonics of the optical driving field are produced for a gas or vapor medium. The substructure that is apparent for $\lambda \gtrsim 22$ nm in the 30 fs spectrum of Fig. 2 is the result of shorter wavelength features appearing in second order. It is also apparent that resorting to <10 fs laser pulses results in discrete harmonics merging into a continuum that extends to the cutoff at (in this case) ~ 8 nm ($\hbar\omega_{MAX} \sim 155$ eV). From these and other experiments cited earlier, it has become evident that reducing the pulse width of the laser driver has proven effective in decreasing the shortest wavelength attainable with HHG, underscoring the conclusion of Ref. [46] that "…efficient production of coherent X-ray photons in excess of 400 eV calls for pulses of 10 fs or shorter in duration."

A qualitative portrayal of one facet of the enormous progress in HHG that was realized in the 1975–2000 time period is given in Fig. 3. For the sake of simplicity, only the highest harmonic order (rather than the maximum photon energy), produced by a given laser wavelength in a particular set of experiments, is presented. Results for CO_2, Nd:glass (or Nd:YAG), KrF (248 nm), and Ti:sapphire laser drivers are shown and the inexorable progression to higher photon energies is clear.

3.1.2. Conversion efficiency of HHG

In any assessment of the potential of HHG as a viable tool for quantum electronics, energy conversion efficiency must hold a prominent position. Early estimates of absolute photon yields by Li et al. [33] with an Nd:YAG system suggested that $\sim 10^5$ photons/harmonic were produced for each of the plateau harmonics ($q \approx 7$–29) in Ar for peak laser pulse intensities (I_p) of 3×10^{13} W cm^{-2}. At higher peak intensities (1.5×10^{15} W cm^{-2}) and $\lambda = 1.053$ μm (Nd:glass), L'Huillier and Balcou [38] reported photon yields for Ar that jumped to $\sim 2 \times 10^7$ for the 30th harmonic and, in the plateau of the Ne spectrum (extending beyond $q = 125$) at this intensity, 10^5–10^6 photons/pulse were generated in each harmonic. Much lower yields ($\sim 10^4$ photons/pulse) were reported in Ref. [39] for Ne photo-excited by Ti:sapphire laser pulses having $I_p > 3 \times 10^{15}$ W cm^{-2}. A comparison of the harmonic photon yields for the lighter rare gases (He, Ne, and Ar), irradiated by either Ti:sapphire or KrF laser pulses, is reprinted from Kondo et al. [39] as Fig. 4. A clear plateau is observed for Ne and the nominal photon yield/harmonic is 10^4 which, for $\hbar\omega = 50$ eV, corresponds to a pulse energy of ~ 0.1 pJ.

A detailed study of the absolute energies produced at particular harmonics was described by Ditmire et al. [49] at the Lawrence Livermore National Laboratory who calibrated the HHG detection system against a well-characterized synchrotron source of XUV radiation, and measured the energy yields and conversion efficiencies for harmonics of an Nd:glass laser (1.053 μm and its second harmonic at 526 nm) in the 17–31 nm spectral interval. Fig. 5 summarizes several of the results of Ref. [49]. For a driving optical field wavelength of 526 nm and $I_p = 10^{15}$ W cm^{-2} (50 mJ, 650 fs, 140 μm spot size), for example, single pulse energies of 1–2 nJ were produced in Ne in the plateau harmonics ($q = 17$–21), an output corresponding to a conversion efficiency of $\sim 2 \times 10^{-8}$. When the fundamental wavelength was 1053 nm, however,

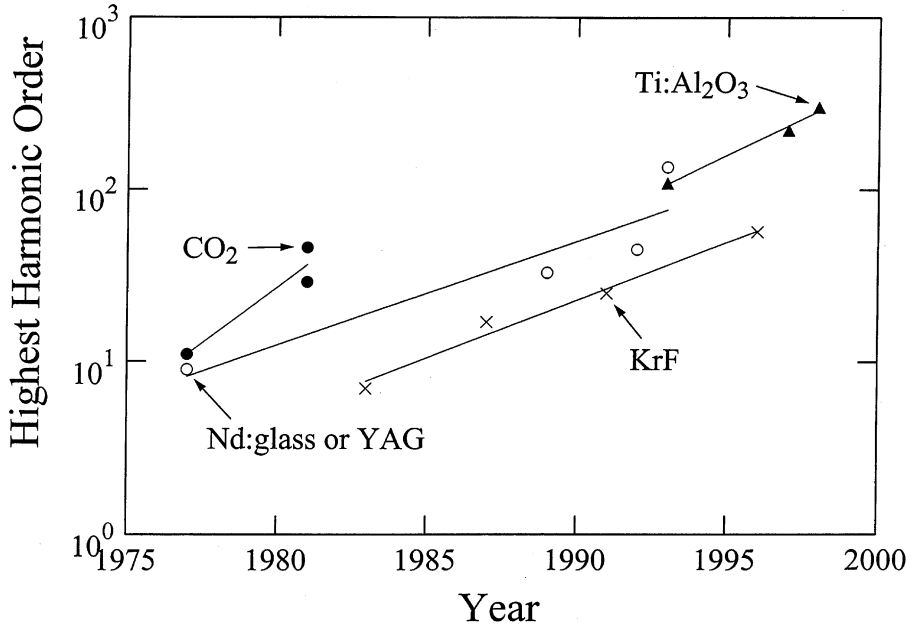


Fig. 3. Several milestones in the progression toward harmonics of ever increasing order in the 1975–2000 time frame. The CO_2 laser data were obtained with solid targets (Refs. [20–23]) whereas the Nd:YAG [or Nd:glass] (Refs. [27, 32, 33, 36, 38]), KrF (Refs. [29, 31, 34, 42]) and Ti:Al_2O_3 (Refs. [37, 43, 46]) laser experiments employed the rare gases as the HHG medium.

the photon yield in the plateau (in Ne) and 6–8 pJ/harmonic fell, resulting in pulse energies of an energy conversion efficiency of $\sim 10^{-10}$.

By optimizing the gas pressure in a hollow fiber (200 μm i.d.), Constant et al. [50] subsequently obtained an energy efficiency of 4×10^{-5} ($\sim 1.5 \times 10^{10}$ photons/pulse) for the conversion of pump energy into the 15th harmonic of 800 nm (Ti:sapphire) in Xe. Although similar measurements were not reported for Ne, the maximum efficiency measured for the same harmonic and laser intensity ($1.5 \times 10^{14}\,W\,cm^{-2}$) but with a jet source of the Xe gas, is approximately a factor of two lower than that for the gas-filled hollow fiber. Furthermore, experiments with Ar produced $> 10^9$ photons/pulse for each odd harmonic between $q = 11$ and 23, again for a peak intensity of $1.5 \times 10^{14}\,W\,cm^{-2}$. Thus, the energy conversion efficiency for the 15th harmonic of Ti:sapphire in the Ar-filled fiber is roughly an order of magnitude lower than the analogous value for Xe. With a self-guided beam geometry (having 300 μm dia. entrance and exit spatial filters), Tamaki et al. [51] increased the conversion efficiency of the 49th harmonic (in the cutoff region for $I = 1.5 \times 10^{15}\,W\,cm^{-2}$) of 800 nm in Ne by a factor of 40. The result was conversion efficiencies up to 10^{-6} and single pulse energies of a few nJ (for $q = 49$, $\hbar\omega \sim 76$ eV). Intensities of several plateau harmonics (such as $q = 25$) were increased by a factor of four.

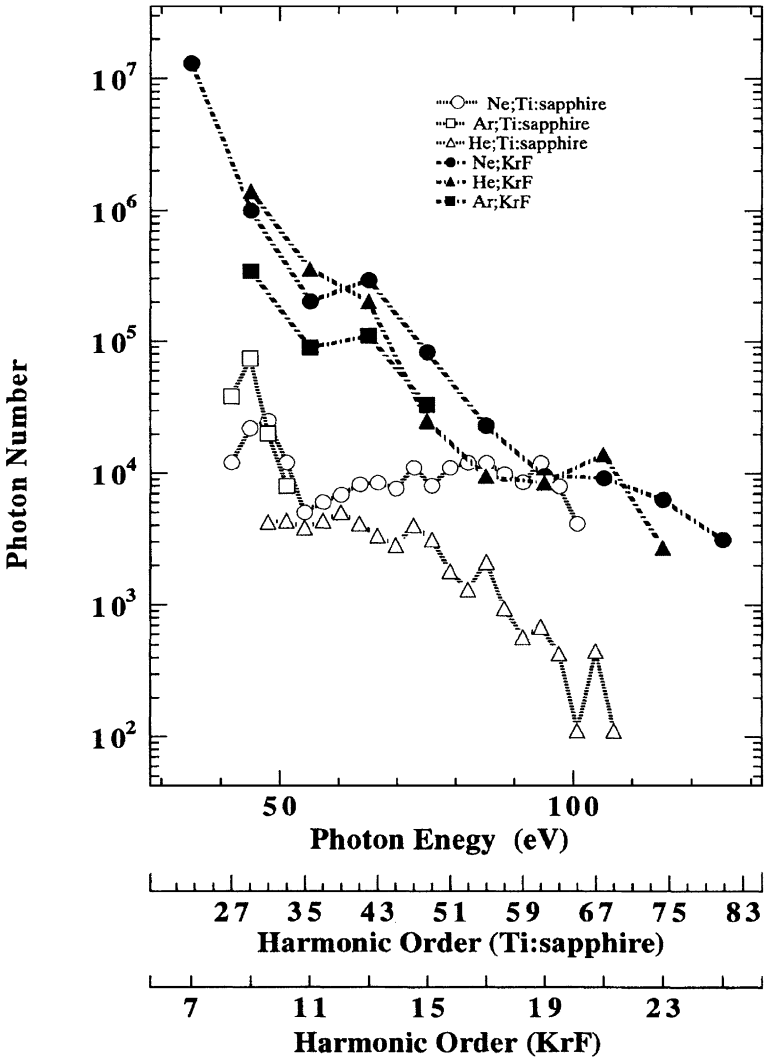


Fig. 4. Comparison of the absolute photon yields for harmonics produced by the Ti:sapphire (~120 fs) and KrF (~500 fs) lasers in He, Ne, and Ar (reprinted from Ref. [39], by permission). For the Ti:sapphire laser data, the peak pulse intensity is $3.2 \times 10^{15}\,W\,cm^{-2}$ for He, $3.1 \times 10^{15}\,W\,cm^{-2}$ for Ne, and $3.3 \times 10^{15}\,W\,cm^{-2}$ for Ar. The rare gas number densities are $1.9 \times 10^{18}\,cm^{-3}$ for He and Ne, and $7.9 \times 10^{17}\,cm^{-3}$ for Ar.

Recently, Poletto, Tondello, and Villoresi [48] measured "global" photon yields — the total number of photons emitted by a group of harmonics — for HHG in He with 30 fs or sub-10 fs pulses from a Ti:Al_2O_3 laser. In the 7–25 nm spectral region ($q = 33$–75), for example, 30 fs pulses focused onto a He jet, for a peak intensity of $3.5 \times 10^{14}\,W\,cm^{-2}$, produced 3×10^7 photons/pulse which corresponds to

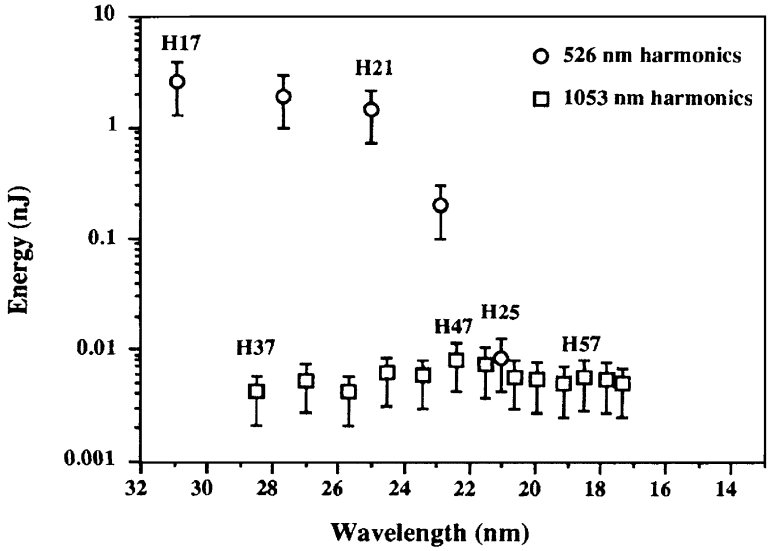


Fig. 5. Energies produced for harmonics 37–61 of Nd:glass laser ($\lambda = 1053$ nm) pulses in Ne. Data for the second harmonic are also shown by the open circles (○). For both sets of experiments, the peak laser intensity was 10^{15} W cm^{-2} (reprinted, by permission, from Ref. [49]).

a global energy conversion efficiency of $\sim 9 \times 10^{-7}$. If the laser pulse width is reduced below 10 fs in otherwise identical experiments, the photon yield drops to 1×10^{7}/pulse and the conversion efficiency is $\sim 6 \times 10^{-7}$, despite the fact that the peak intensity increased to $\sim 9 \times 10^{14}$ W cm^{-2}. Subsequent experiments by Takahashi and co-workers [52–56] with self-guided beams or weak-focusing of the driving optical field have achieved single pulse energies and conversion efficiencies of ~ 1 nJ and 8×10^{-9}, respectively, for the 81st harmonic ($\hbar\omega = 126$ eV) in He, ~ 25 nJ and $\sim 5 \times 10^{-7}$, respectively, for the 59th harmonic ($\lambda = 13.5$ nm, $\hbar\omega \cong 92$ eV) in Ne, and ~ 0.75 μJ and $\sim 1.4 \times 10^{-5}$, respectively, for the $q = 33$–65 harmonics (combined) in Ne. In the XUV, 10 μJ pulses were reported in Ref. [53] for the $q = 11$–15 harmonics as a group (54–72.7 nm) in Xe, and single harmonic pulse energies and conversion efficiencies as high as 7 μJ and 4×10^{-4}, respectively, were obtained for the 11th harmonic.

Although additional experimental studies of the photon yields into specific harmonics for a particular gaseous medium (the lighter rare gases, in particular) and over a wide range in intensity and focusing conditions (confocal parameter) are clearly warranted, it is fair to say that energy conversion efficiencies $> 10^{-7}$ for a single harmonic in the plateau are currently attainable with sub-50 fs laser pulses and a gas jet target. If the rare gas is entrained in a hollow fiber or a self-guided beam geometry is employed, an improvement of at least an order-of-magnitude in conversion efficiency can be expected. However, further comments regarding the photon yields/harmonic obtainable with hollow, or

photonic fibers and their promise for realizing a practical and compact HHG radiation source in the VUV/XUV will be deferred until Sections 3.1.5 and 3.1.6.

3.1.3. Temporal, spatial, spectral, and polarization characteristics of harmonic radiation

Although reductions in the pulse width of the optical driving field have proven to be effective for influencing the spectral distribution of the harmonics generated and the photon yield, a number of potential applications of HHG are more dependent upon other characteristics of the harmonic radiation, including temporal and spatial coherence, and polarization properties. An extensive experimental effort has been devoted in the past decade to detailing the characteristics of harmonic radiation and is briefly reviewed in this section.

3.1.3.1. Harmonic emission pulse widths. The predictions of early calculations [57] of HHG temporal profiles, suggesting that the *pulse widths* of individual harmonics might be considerably narrower than that for the fundamental, were confirmed by Schins et al. [58] in 1996. In cross-correlation experiments with a 150 fs, 800 nm (Ti:Al_2O_3 laser) pulse which was time-delayed with respect to the 21st harmonic of the fundamental wavelength, measurements of photoelectron spectra suggested a duration of 40 fs for the harmonics generated. Similar measurements by Glover and co-workers [59] yielded a pulse duration of 50 ± 15 fs for the 23rd harmonic of 800 nm for 70 fs driver pulses. Bouhal et al. [60] also adopted a cross-correlation approach to find that the pulse widths of the 21st and 27th harmonics of 800 nm in He are (to within experimental uncertainty) both 130 ± 30 fs, for a driving optical pulse duration of 190 fs (FWHM). Bellini and co-workers [61] applied Michelson interferometry to measuring the temporal coherence associated with harmonics generated in Ar. With 100 fs pump pulses at 790 nm, coherence times measured for most of the harmonics investigated ($q = 7$–21) suggest harmonic pulse widths of 20–40 fs, a result in agreement with the conclusions of Ref. [57]. All of these results are consistent with the theoretical expectation [62] that "…harmonics are created only during the rising edge of the fundamental pulse" [58]. Theoretical and experimental studies [40, 47] in the intervening years have generally upheld this view and the recent successful generation and measurement of sub-fs (attosecond) pulses [63], to be addressed in Section 3.1.4, exploited this and other characteristics of HHG. Phase-locking harmonics, produced by 40 fs laser pulses, was found to generate trains of ~ 250 as duration (FWHM) [63] pulses.

3.1.3.2. HHG spatial characteristics. Measurements of the *spatial properties* of harmonics in the plateau and above cutoff were reported by Tisch et al. [64] for He and driver intensities (at $\lambda = 1.05\,\mu$m, 1.3 ps pulse width) up to $3 \times 10^{14}\,\mathrm{W\,cm^{-2}}$. Plateau harmonics displayed considerable structure but the transverse intensity distribution approached a Gaussian when above cutoff. For $q = 111$, the beam divergence was determined to be < 2 mrad.

Measurements by Peatross and Meyerhofer [65, 66] of the far-field intensity patterns associated with the $q = 11$–41 harmonics of 1.054 μm (1.4 ps pulse) in Ar, Kr, and Xe at low pressure (<1 Torr) revealed a narrow central peak having a divergence (FWHM) of <10 mrad. However, wings containing ~75% of the total pulse energy for that harmonic were also observed. Salières and co-workers [67] found no substructure in the harmonic radiation cross-sectional profiles with shorter (140 fs) laser pulses and Ne or Ar as the conversion medium. For all but the highest intensities investigated ($\sim 4 \times 10^{14}$ W cm^{-2}), the transverse intensity profiles were similar: the divergence of plateau harmonic radiation was typically 15 mrad, but declined to 6 mrad (or ~3× larger than that observed in Ref. [64]) above cutoff. Fig. 6 displays the transverse intensity profiles for several harmonics of the Cr:$LiSrAlF_6$ laser (825 nm), generated in Ar and Ne and described in Ref. [67]. Both rare gas media exhibit significant narrowing for the higher-order harmonics. The divergence of the 29th harmonic in Ar (upper portion of Fig. 6), for example, is less than half that of the 23rd harmonic. More recent experiments by Salières et al. [68], also conducted with a fundamental optical field wavelength of 825 nm (Cr:$LiSrAlF_6$; 280 mJ/pulse, 1 Hz), confirmed that the plateau harmonics in Ne and Ar exhibit similar beam divergences (~14 mrad for $q = 31$–37 in Ne for $I_p = 2.7 \times 10^{14}$ W cm^{-2}, for example) [68], whereas, in the cutoff region, the divergence of the harmonic radiation declines sharply with increasing harmonic order (for a fixed value of I_p). The latter bodes well for applications since energy/pulse is generally at a premium for harmonics above cutoff and beam quality is essential for maximizing focal intensity. The experiments of Ref. [68] also graphically demonstrated the influence of the position of the pump focus relative to the gas medium (gas jet in Ref. [68]). Optimizing harmonic beam quality while maintaining reasonable energy conversion efficiencies requires situating the gaseous medium *after* the laser focal plane. Spielmann et al. [47] have characterized, by a knife edge measurement technique, the coherent XUV continua produced in He with still shorter pulses (5–7 fs), and measured e^{-2} beam widths of 480 and 180 μm at wavelengths of 10 and 4.4 nm, respectively. These values correspond to a beam divergence within a factor of three of the diffraction limit!

More recent measurements of beam divergence by Takahashi et al. [54, 55] confirm the near diffraction-limited quality of higher harmonic radiation. The far-field spatial profiles for the 59th ($\lambda = 13.5$ nm) and 81st (9.8 nm) harmonics, for example, produced in Ne and He, respectively, at an intensity of $\sim 4 \times 10^{14}$ W cm^{-2} were shown to be near-Gaussian and correspond to divergences of 0.35 and 0.3 mrad (respectively).

Experiments suggest, therefore, that the substructure or "wings" observed in the intensity profiles of harmonics generated with longer pulse ($\gtrsim 1$ ps) excitation are absent when the laser pulse widths are decreased to, or below, ~100 fs. As suggested in Ref. [64], the free electrons produced by photoionization of the gaseous medium before peak laser intensity is reached are undoubtedly responsible for this phenomenon.

These conclusions are supported by studies of the spatial coherence of harmonics produced in He with 2 ps, 1.054 μm (Nd:glass) laser pulses. Young's double slit experiments reported by Ditmire and co-workers [69] showed (Fig. 7) that the fringe

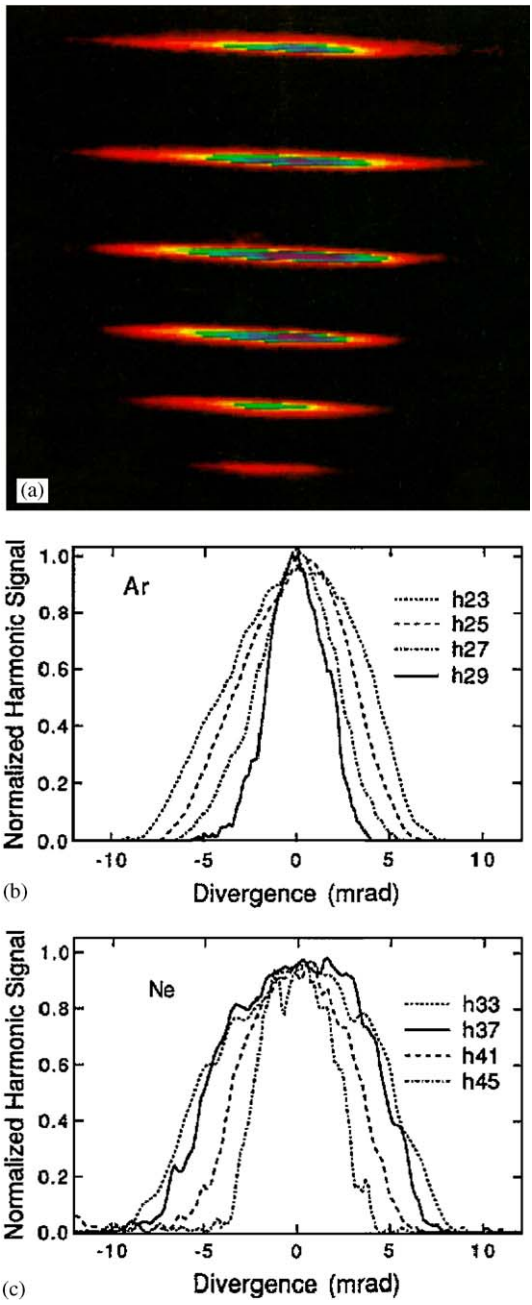


Fig. 6. Cross-sectional intensity profiles for the harmonics of a Cr:LiSrAlF$_6$ laser ($\lambda = 825$ nm): (top) $q = 23$–29 in Ar for a peak intensity of 1.8×10^{14} W cm^{-2}; (bottom) $q = 33$, 37, 41, and 45 produced in Ne at a peak intensity of 2.7×10^{14} W cm^{-2} (reprinted by permission from Ref. [67]).

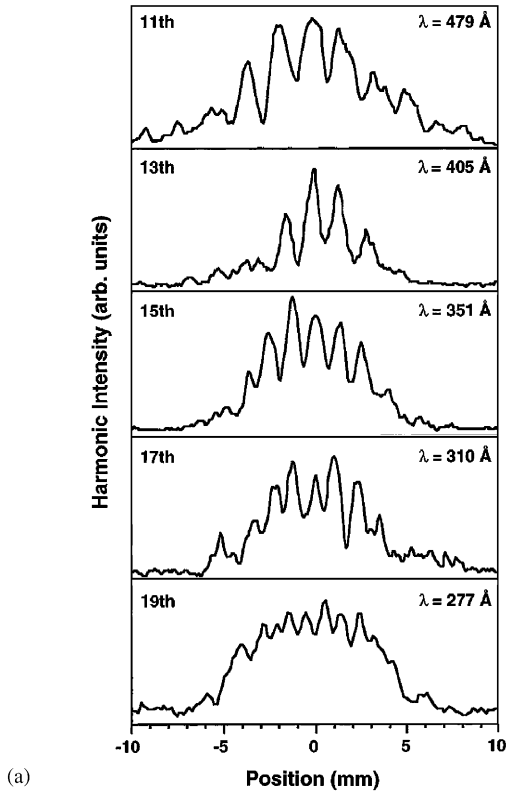


Fig. 7. Measurements of fringe visibility (a measure of spatial coherence) for harmonics generated in He with an Nd:glass laser (1.054 μm, 2 ps pulses): (a) Spatial scans of the intensity patterns produced in a Young's double slit experiment for $q = 11$–19; (b) measured fringe visibility for the 11th–19th harmonics for two values of peak laser intensity (1×10^{15} and 4×10^{15} W cm^{-2}). Reprinted by permission from Ref. [69].

visibility decreases with both harmonic order number and peak laser intensity. For the eleventh harmonic (47.9 nm), the fringe visibility for $I_p = 10^{15}\,W\,cm^{-2}$ is ~ 0.6 which indicates that the emission is partially coherent. It should be noted that this degree of spatial coherence is comparable to that for the Ne-like Ar discharge amplifier at 46.9 nm which was found to be 0.6 for a capillary plasma column length of 10.6 cm (Ref. [70]). Fringe visibilities up to $\sim 30\%$ were observed by Zerne et al. [71] for harmonics generated in Xe ($q = 7$–17) by longer pulses (35 ps) at 1.064 μm. With a clever approach to producing two harmonic sources, separated by 150 μm in the same gas jet, they were able to demonstrate conclusively that the spatial and temporal coherence of the 9th–13th harmonics, although diminished relative to that for harmonics produced by shorter pump pulses, is nevertheless ample to phase-lock the two sources. For a detailed review of the spatial and temporal coherence of HHG radiation and the predictions of semiclassical theory, the reader is referred to Ref. [72].

3.1.3.3. Spectral linewidths. In recent experiments with both 30 and 7 fs laser pulses (fundamental wavelength of 795 nm), Villoresi et al. [73] measured the *linewidths* of harmonics generated in Ne in the ~ 90–260 Å region ($q \approx 31$–87) and observed markedly different behavior for the two pulse widths. As illustrated in Fig. 8 (reprinted from Ref. [73]), the linewidths of harmonics produced by 30 fs pulses

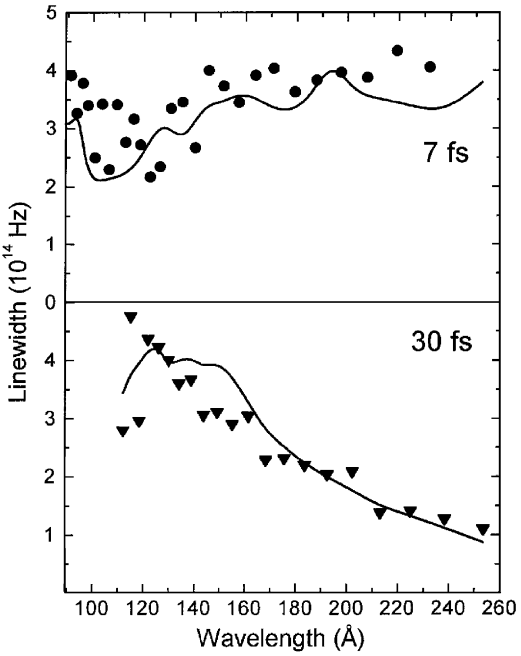


Fig. 8. Linewidths of harmonics in the ~ 90–260 Å region ($q \approx 31$–87) generated in Ne by 7 fs or 30 fs laser pulses at 795 nm. The solid curves in both graphs represent the predictions of computer simulations (reprinted from Ref. [73], by permission).

(lower panel, Fig. 8) rise steadily with increasing harmonic order. Harmonics in the plateau exhibit linewidths of $\sim 10^{14}$ Hz whereas those above cutoff are considerably broader ($\Delta\nu \sim (3\text{–}4) \times 10^{14}$ Hz). In contrast, the harmonics resulting from 7 fs pulses are characterized by linewidths that vary slowly with harmonic order number and, in fact, a slight *decrease* in spectral width is observed for the shortest wavelength harmonics.

3.1.3.4. Polarization properties. A critical property of HHG is, from several perspectives, the *polarization* of the harmonics. Not only is the harmonic response to laser pulses of variable ellipticity of value for altering the beam properties, but the polarization behavior of HHG provides window into the nature of the harmonic generation mechanism itself. As Antoine et al. [74] have stated succinctly: "The recognizable features of harmonic spectra, in particular the extended plateau followed by the sharp cutoff, are predicted by many theoretical models. More severe tests are required to differentiate among them and hence to improve the understanding of the process. Polarization measurements provide such a test." For these reasons, a number of experimental studies of the effect of fundamental field ellipticity on harmonic generation have appeared in the literature [23, 74–85]. Early experiments by Carman, Rhodes, and Benjamin [23] with a fundamental wavelength of 10.6 μm (CO_2 laser) showed the visible harmonics to be linearly polarized, with 90% of the harmonic radiation polarized parallel to the driving (fundamental) optical field. Subsequent studies of the polarization properties of harmonic radiation fall into two broad categories. The first (and most commonly reported measurement) is concerned with the variation of the relative harmonic yield with the degree of fundamental field ellipticity [66, 75–79, 83–85] whereas the second addresses the characteristics of the polarization ellipse associated with a given harmonic [74, 80, 82]. Measurements reported by Budil and co-workers [75] in 1993 exhibited a rapid decline in harmonic generation efficiency as the fundamental field ellipticity was increased. A sample of the experimental results of Ref. [75] for several harmonics between $q = 15$ and 63 is shown in Fig. 9 for Ne as the conversion medium and a fundamental wavelength of 825 nm. Notice that ellipticities of only 0.1–0.2 are sufficient to reduce the photon yield by a factor of two. It is also clear that the impact of rising ellipticity in the fundamental optical field on the harmonic yield is increasingly pronounced for the higher harmonics. Similar observations were reported for other rare gases and fundamental wavelengths [75–79, 81, 83–85].

Another diagnostic of the polarization properties of harmonic radiation is the behavior of the polarization ellipse for each harmonic. For a given ellipticity of the fundamental optical field, the harmonic radiation will, in general, also be elliptically polarized and its polarization ellipse may have an ellipticity different from that for the fundamental *and* be rotated relative to the ellipse associated with the fundamental. The first experiments in this area were published by Weihe et al. [80, 82], who measured the ellipticity of the harmonic radiation as well as the offset (or rotation) angle — the angle between the major axis of the harmonic's polarization ellipse and that for the fundamental. Measurements were made in both Ar and N_2 as a function of the ellipticity of the fundamental field and, in Ref. [86], offset angle

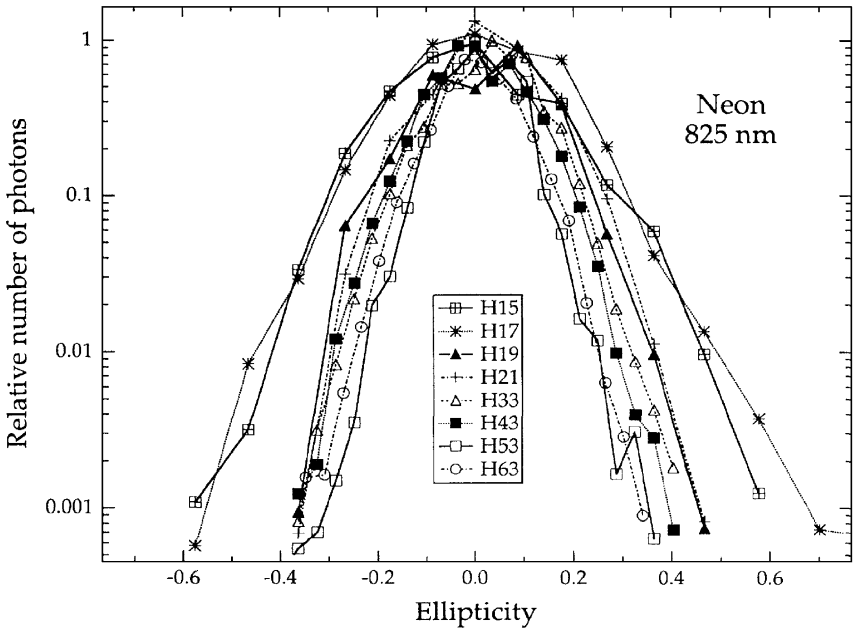


Fig. 9. Relative photon yield (number of photons generated/pulse) for eight harmonics of 825 nm in Ne. The peak fundamental intensity and pulse width are $10^{15}\,W\,cm^{-2}$ and 140 fs, respectively (reprinted from Ref. [75], by permission).

data for the 7th–13th harmonics of 785 nm in Ar were compared with the predictions of a quantum electrodynamic theory (to be described in Section 4.2).

Extensive measurements of the rotation angle and ellipticity of the harmonics of 790 nm generated in Ne, Ar, and Xe were reported by Antoine et al. [74] in 1997. Acquired for a range of harmonics in both the plateau and cutoff regions of the spectrum, these data illustrate the complexity of characterizing in detail the polarization characteristics of HHG radiation. Fig. 10, for example, presents experimental results from Ref. [74] in which the rotation angle of the polarization ellipse has been measured as a function of fundamental ellipticity for the $q = 17$, 23, 29, and 33 harmonics of 790 nm (Ti:Al_2O_3) in Ne for fundamental intensities of $(4–6) \times 10^{14}\,W\,cm^{-2}$. For a given value of fundamental ellipticity, the rotation angle clearly declines as the harmonic order number increases. Given a fundamental ellipticity of 0.3, for example, the rotation angle for the polarization ellipse of the lowest harmonic studied ($q = 17$) is 40°! Of course, the rotation angle data (such as those of Fig. 10) provide only one perspective of the HHG process and data summarizing the measurements of harmonic ellipticity as a function of the fundamental ellipticity are reprinted from Ref. [74] in Fig. 11. The 17th, 23rd and 29th harmonics of 790 nm in Ne are generated with fundamental ellipticities as high as ~ 0.3 and the crucial observation is that, particularly for the lower harmonics, the harmonic ellipticity rises more quickly with increasing fundamental ellipticity than

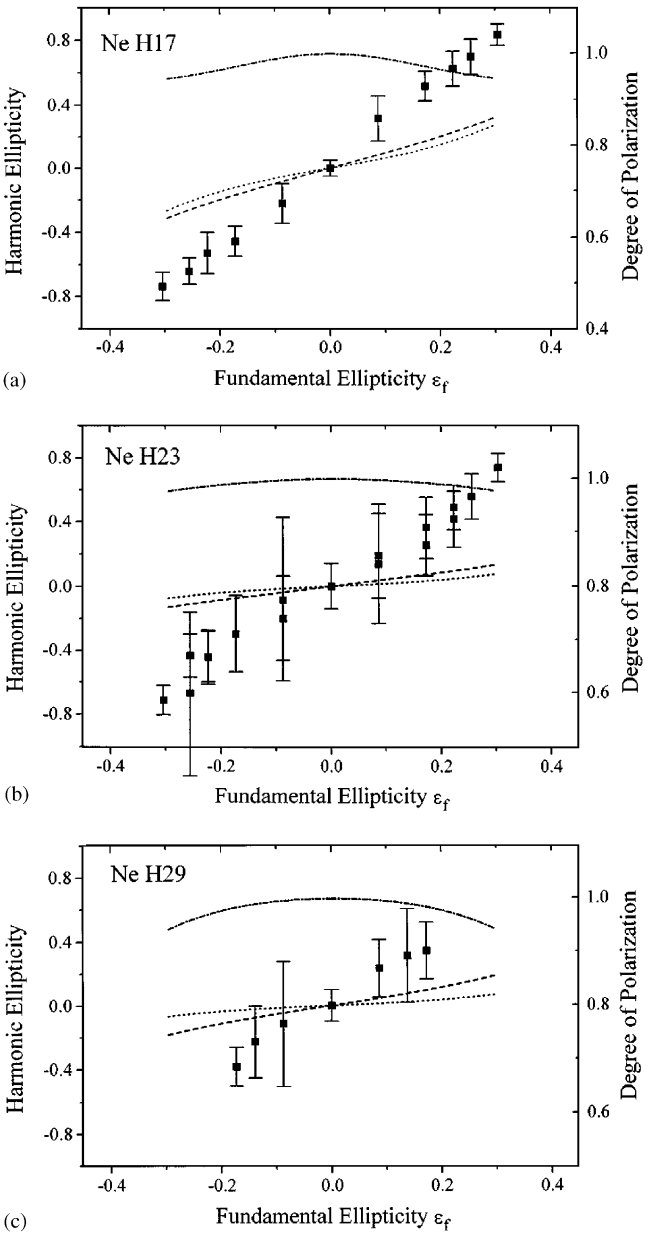


Fig. 10. Polarization data of Antoine et al. (Ref. [74]), illustrating the variation of the rotation angle (of the harmonic polarization ellipse) with the ellipticity of the fundamental field (reprinted from Ref. [74], by permission). The results shown are for the $q = 17$, 23, 29, and 33 harmonics in Ne at an intensity of $(4–6) \times 10^{14}\,\mathrm{W\,cm^{-2}}$. The dashed curves represent the predictions of a semiclassical model of the HHG process, which will be discussed in Section 4.1.

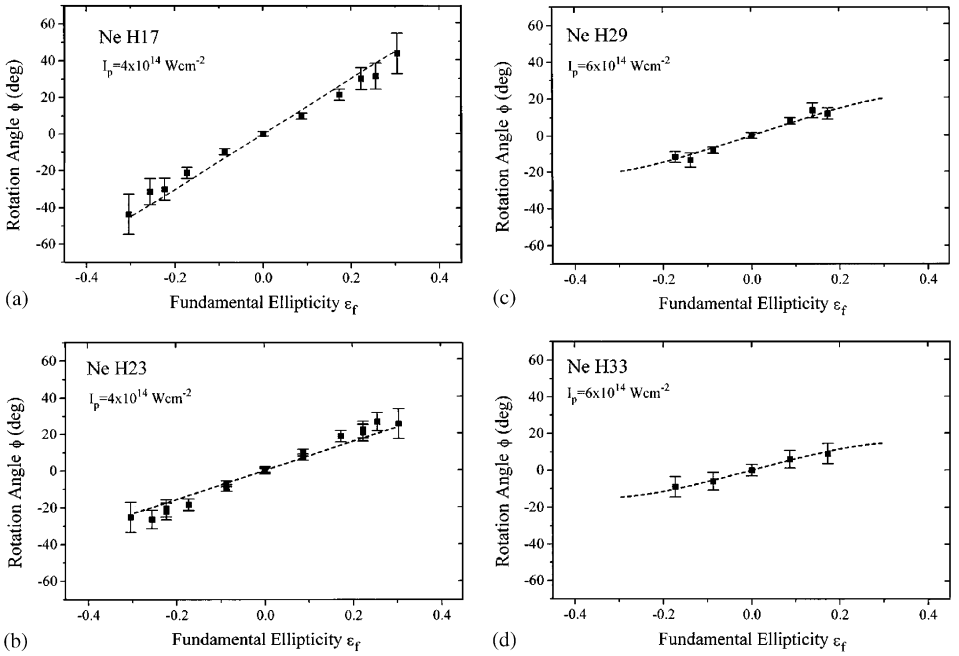


Fig. 11. Ellipticity data for the $q=17$, 23, and 29 harmonics of 790 nm (Ti:sapphire) in Ne, showing the dependence of the harmonic ellipticity on the ellipticity of the fundamental. The dashed (and dotted) curves again represent the predictions of semiclassical theory (reprinted from Ref. [74], by permission).

expected. Considering, for example, the results for $q=17$ shown in Fig. 11(a), it is clear that the harmonic ellipticity varies (to within experimental uncertainty) linearly with the fundamental ellipticity over the range investigated. The slope of the linear least-squares fit to the data is ~ 3, and, thus, the ellipticity of the 17th harmonic is approximately triple that for the fundamental. Although this effect weakens with increasing harmonic order (and the heavier rare gases) [74], its magnitude is surprising but of value when developing optical systems and applications for harmonic radiation. In general, experiments show that the rotation angle for the polarization ellipse is large for harmonics in the plateau and the ellipticity is similar to that for the fundamental. In the cutoff region, however, the rotation angle and ellipticity for the harmonics are small and essentially independent of the fundamental ellipticity. Mercer et al. [87] have demonstrated that the dependence of harmonic generation efficiency on pump ellipticity can be exploited to control the spatial mode structure (and corresponding divergence) of individual harmonics.

One final characteristic of HHG radiation that should be mentioned is the blue shift of harmonics [37, 57, 88–92] that is a direct outgrowth of the temporal history of the electron density $\left(\text{i.e.,}\dfrac{\partial n_e}{\partial t}\right)$ of the medium. Experiments [89] characterizing the third harmonic of ~ 795 nm in the rare gases showed that for maximum

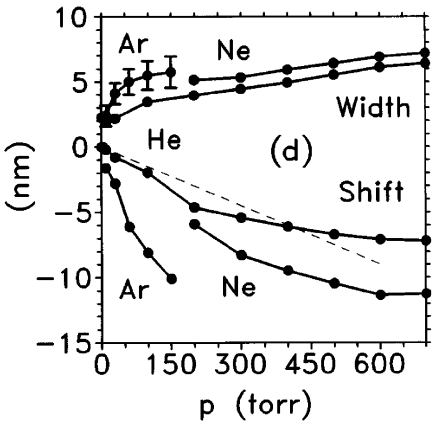


Fig. 12. Data reprinted from Ref. [89] (by permission) of the shift and broadening of the third harmonic of $\lambda \sim 795$ nm in He, Ne, and Ar as a function of the rare gas pressure.

instantaneous intensities of $\sim 10^{16}\,\mathrm{W\,cm^{-2}}$, the blue shift of the peak wavelength of the harmonic ranges from $\Delta\omega/\omega = 10^{-5}$–$10^{-1}$ for gas pressures of 0.1–700 Torr, respectively. A portion of the data reported in Ref. [89] is illustrated in Fig. 12 (reprinted by permission) and it is clear that varying the pressure of the gas medium offers a significant degree of tunability ($\lesssim 11$ nm), although one must also be prepared to accept an increase in the bandwidth of the harmonic (as large as a factor of two). Adopting a different approach, Altucci et al. [90] subsequently found that continuous tunability in the spectral region below 20 nm is feasible. Blue shifts of 0.3–0.4 nm for the plateau harmonics were observed when the position of the pump focal point relative to a Ne gas jet, rather than the gas pressure, was varied for peak laser intensities of $(2.4\text{–}4.9) \times 10^{14}\,\mathrm{W\,cm^{-2}}$. Shin and co-workers [91, 92] confirmed the degree of the blue shift of the $q = 21$–43 harmonics of 813 nm in Ne and Ar for peak fundamental intensities as high as $1.3 \times 10^{16}\,\mathrm{W\,cm^{-2}}$. Spectra recorded as a function of peak laser intensity for two values of fundamental pulse duration — ~ 30 fs and 80 fs — also displayed the large shifts in the frequency of high harmonics that can be realized. Representative data from Ref. [92], HHG spectra acquired with 27 fs laser pulses ($\lambda = 813$ nm) irradiating Ne at focal intensities of (a) $1.3 \times 10^{16}\,\mathrm{W\,cm^{-2}}$, and (b) $5 \times 10^{15}\,\mathrm{W\,cm^{-2}}$ for a peak Ne number density of $3 \times 10^{18}\,\mathrm{cm^{-3}}$, are presented in Fig. 13. As discussed by Shin et al. [92], the magnitude of the blue shift rises with increasing harmonic order. For example, the wavelength of the 39th harmonic of ~ 813 nm in Ar at a peak laser intensity of $9 \times 10^{15}\,\mathrm{W\,cm^{-2}}$ is the same as that for the *unshifted 41st harmonic*. This corresponds to a shift of twice the fundamental optical field frequency! More impressive still is the observation that the 85th harmonic in Ne for $I_p = 1.3 \times 10^{16}\,\mathrm{W\,cm^{-2}}$ is blue shifted from its expected position by four times the fundamental laser frequency. The surprisingly large magnitude of this phenomenon is, as noted in Ref. [92], "…enough to cover the

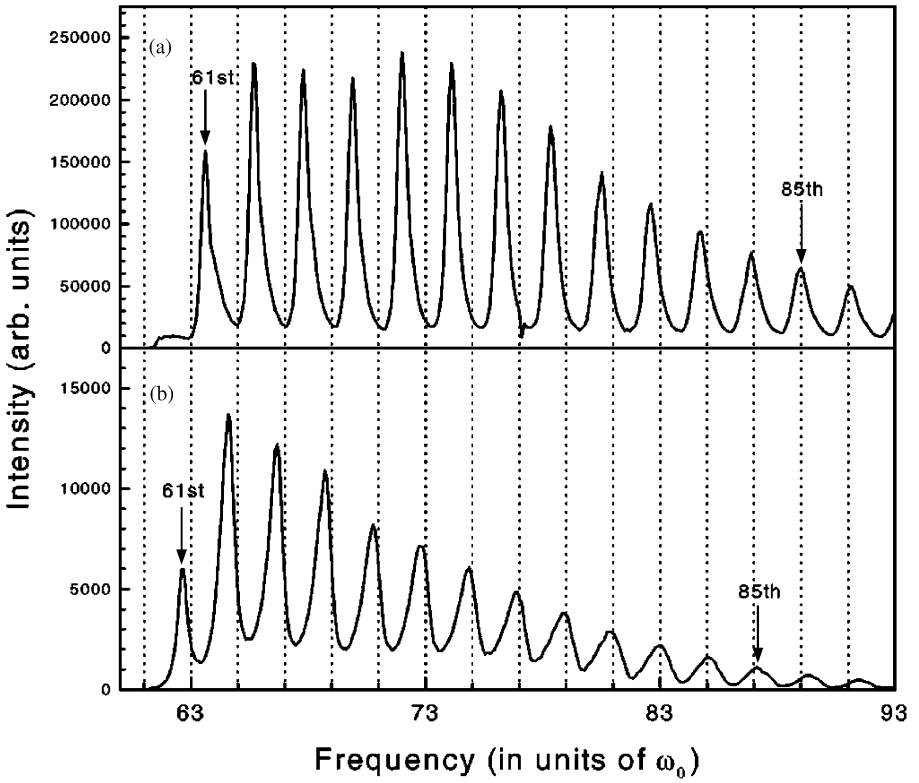


Fig. 13. HHG spectra produced when Ne is irradiated by 27 fs pulses having peak intensities of: (a) $1.3 \times 10^{16}\,\mathrm{W\,cm^{-2}}$ and (b) $5 \times 10^{15}\,\mathrm{W\,cm^{-2}}$ (reprinted from Ref. [92], by permission).

interval between odd harmonics," and was attributed to a nonadiabatic effect rooted in the short risetime of the electric field strength of the driving optical pulse. One concludes that continuous tunability throughout the deep-XUV region and into the soft X-ray ($\lambda \sim 8$–20 nm) realm is now available to the optical scientist or engineer with HHG.

3.1.4. Attosecond pulse generation

The extended bandwidth afforded by a group of harmonics having the proper phase relationship to each other is responsible for a recent milestone in optical physics — crossing the 1 fs pulse duration barrier or, to borrow the phrase of Refs. [93, 94], the "attosecond frontier." Since the early 1990s, theoretical arguments have been advanced [95–97] to describe conditions under which HHG might be controlled so as to produce sub-fs optical pulses. Various approaches, including tailoring the ellipticity of the excitation pulse or driving the HHG process with a single cycle of the fundamental optical field, were proposed but it was not until 1999 that Papadogiannis et al. [98] published evidence for a train of sub-fs pulses and a

single temporal feature having a pulsewidth <100 attoseconds (as). Then, in 2001, a succession of experiments conclusively demonstrated that optical pulses on the attosecond time scale had been realized. Drescher et al. [93] inferred X-ray pulse widths of $1.8\begin{pmatrix}+0.7\\-1.2\end{pmatrix}$ fs for HHG in Ne with 7 fs excitation pulses and, later that year, the same group reported [99] the generation of soft X-ray ($\lambda\sim14$ nm) pulses having durations of 650 ± 150 as. By also adopting the cross-correlation, photoelectron spectroscopy technique of Ref. [58], Paul and co-workers [63] succeeded in measuring the relative phases of the $q=11$–19 harmonics generated in Ar and found that a train of phase-locked pulses, each having a duration of 250 as, was being produced despite the fact that the fundamental optical field ($\lambda=800$ nm) pulsewidth was 40 fs. Other experiments [100], combined with semiclassical theoretical considerations, also suggested that attosecond pulse trains were produced under specific focusing conditions designed to optimize phase matching. It is evident that HHG readily provides the enormous bandwidth required to generate sub-fs optical pulses and that, under proper experimental conditions (focusing geometry, gas pressure, pump-medium interaction length, etc.), the harmonics will, indeed, phase-lock to produce trains of pulses having durations no longer than a few hundreds of attoseconds. As the facility with which the phase of harmonic radiation can be accurately and reproducibly controlled becomes increasingly sophisticated over the next several years, it will undoubtedly become possible, as noted by Paul et al. [63] "…to properly phase all harmonics emerging from the generation jet, leading to pulses perhaps as short as 10 as."

3.1.5. *Experimental considerations*

From a conceptual standpoint, the experimental apparatus required for generating harmonic radiation is straightforward and Fig. 14 is a representative system, drawn from Ref. [9], illustrating the essential elements. Pulses, typically <100 fs in duration, from one or more lasers are focused into a gaseous medium in the form of a jet (produced by a pulsed valve) [101], or one or more gases at a static pressure and contained within a hollow fiber. Although not specifically indicated in Fig. 14, the position of the laser focal plane (denoted $z=0$) relative to the gaseous nonlinear medium has a profound bearing on phase-matching and, therefore, the spectral and spatial characteristics of the radiation emerging from the jet. Owing to phase-matching and the influence of Gouy phase, in particular, optimal harmonic generation, as defined by spectral brightness ($\mathrm{W\,sr^{-1}\,cm^{-2}}$), occurs near the focal plane ($z=0$) but slightly upstream (i.e., $z\lesssim0$). With 7 fs pulses having a peak intensity of $9\times10^{14}\,\mathrm{W\,cm^{-2}}$, for example, spectral brightness values exceeding $6\times10^{13}\,\mathrm{W\,sr^{-1}\,cm^{-2}}$ have been obtained for the $q=41$–71 harmonics of ~800 nm produced in a Ne jet [102]. Higher brightness values have been observed for lower harmonics (see, for example, Ref. [116]).

Of overriding importance to the HHG process is the degree of control over the key parameters of the laser pulses. Chirp, spectral content and the temporal width and profile of the pulses are the primary determinants of the HHG spectrum produced as

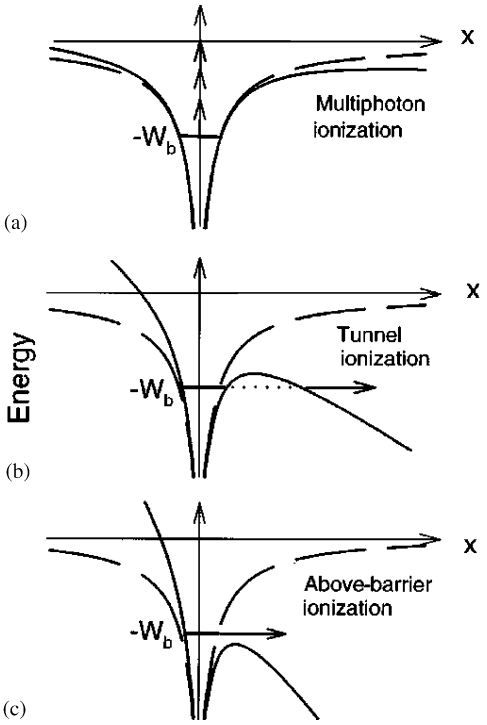


Fig. 14. Schematic diagram of a representative experimental apparatus for generating and detecting harmonic radiation (reprinted from Ref. [9], by permission). In this system, the backing pressure for the rare gas jet is ~ 2.5 atmospheres whereas the base pressure for the vacuum system is typically $< 10^{-5}$ Torr.

well as the overall efficiency for the process. As discussed in Section 3.1.3, driving HHG with the shortest possible laser pulses is essential for producing photons in the XUV or soft X-ray regions with maximum efficiency. Toward that end, Nisoli et al. [103, 104] introduced in 1996 the gas-filled hollow fiber to serve simultaneously as the HHG nonlinear medium and as a pulse compression device. With a fiber having an inner diameter and length of $\sim 160\,\mu$m and > 60 cm, respectively, and rare gas pressures of several atmospheres, it was shown that spectral broadening up to 250 nm and transmissions through the fiber exceeding 60% could be achieved. As an example of the spectra generated by self-phase modulation (SPM) in a gas-filled hollow fiber, Fig. 15(a) presents the spectrum recorded from a 60 cm long fiber (160 μm i.d.) pressurized with Kr to 2.1 bars. For an input pulse peak power of 2×10^9 W ($\lambda = 780$ nm), the spectrum of the radiation emerging from the fiber is broadened to > 120 THz which is sufficient to yield pulse durations of ~ 4.5 fs as shown by the interferometric autocorrelation trace in the lower panel of Fig. 15. Durfee et al. [105, 106] reported that 20 fs, 800 nm pulses compressed in a 140 μm i.d. fiber, 70 cm in length and back-filled with ~ 70 Torr Ar, yielded 8 fs FWHM pulses at 270 nm (third harmonic) in a parametric process [107–109] in which the second harmonic (400 nm) and fundamental served as the pump and idler, respectively. The

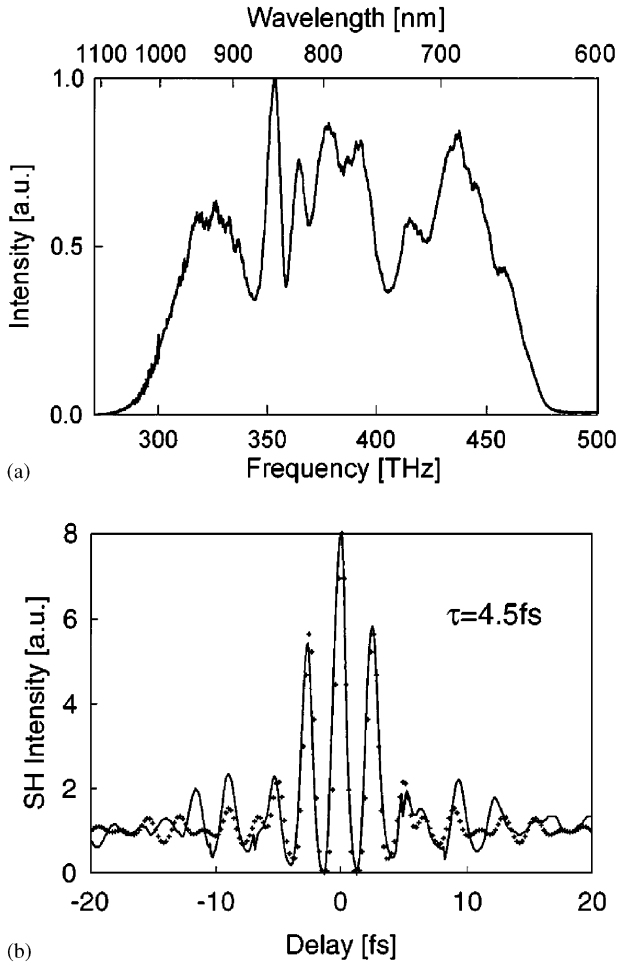


Fig. 15. (a) Near-infrared spectrum resulting from broadening of 20 fs, 300 μJ pulses (centered at 780 nm) in 2.1 bars of Kr. The rare gas is contained within a hollow, fused silica fiber having an inner diameter of 160 μm; (b) interferometric autocorrelation trace associated with compression of the spectrum of part (a), yielding a pulse width (FWHM) of ~4.5 fs (reprinted by permission from Ref. [104]).

energy conversion efficiency for the UV output with respect to the pump pulse was measured to be 16%. Consequently, hollow fibers are effective in a dual role. Not only is one able to temporally compress sub-50 fs pulses but the energy conversion into one or more harmonics can be enhanced by phase matching in a nonlinear medium of controllable length and pressure in which the propagating pump radiation can be confined to the lowest order mode (EH_{11}) [105, 106]. Significant enhancements in the energy conversion efficiency for higher harmonics have also been observed with gas-filled fibers. With 7 fs laser pulses having peak intensities of $(5–10) \times 10^{14}\,W\,cm^{-2}$, efficiencies for converting the 790 nm-centered fundamental

into the $q = 17$–31 harmonics has been measured to be as high as $\sim 10^{-5}$ for an Ar-filled fiber which is more than an order of magnitude larger than the corresponding values attainable with a Ne gas jet [110]. Finally, structuring the inner wall of a hollow fiber such that its inner diameter is modulated with a periodicity of 100 μm–1 mm has been shown [111–113] to further improve phase-matching of the fundamental (pump) with respect to the high harmonics, particularly when the generation of < 10 fs harmonic pulses is desired. With a 25 fs laser system and a 150 μm i.d. hollow fiber in which the bore was modulated by $\sim 13\%$ with a periodicity of 0.5–1 mm, Paul et al. [112] were able to extend the photon energy range over which quasi-phase matching of HHG has been achieved to beyond 100 eV, and the brightness of harmonics in this spectral region was increased by more than two orders of magnitude. A dramatic demonstration of the phase matching of HHG in the water window (4.4 nm, $\hbar\omega \simeq 284$ eV) was reported recently by Gibson et al. [113] who also employed a hollow fiber with a nominal i.d. of 150 μm but modulated with a periodicity of 250 μm. With only 9 Torr of Ne in the fiber, the harmonic flux was increased by a factor as large as three with respect to a straight (unmodulated bore) fiber and the peak intensity at $\hbar\omega = 250$ eV rose by roughly a factor of five. Also, Homoelle et al. [114] have improved the phase contrast of 60 fs pulses by several orders of magnitude with a gas (Xe)-filled hollow fiber placed before the pulse compressor. In summary, although several improvements to the gas jet have been reported, including cryogenic cooling [115], length extension with a slit nozzle configuration [116], and atomic cluster formation in the jet [117], the hollow fiber consistently offers superior performance in terms of output beam quality [118] (< 1 mrad for $q = 17$–23 in Ar) and energy conversion, and is the preferred device for producing harmonic radiation from sub-50 fs fundamental pulses.

Another critical aspect of the fundamental field optical pulses is chirp which influences both the temporal width of the harmonic radiation pulses as well as the HHG spectrum itself. Lee et al. [119] vividly illustrated the impact of positive or negative chirp on the resulting harmonic distribution and, by intentionally introducing positive chirp by placing a LiF disc into the path of ~ 115 fs pulses, Sekikawa and co-workers [120] were able to obtain temporally-compressed pulses having durations as short as 13 fs.

An order of magnitude improvement in the peak intensity of specific harmonics in the X-ray region was reported recently by Bartels et al. [121] with an adaptive optics system designed specifically to tailor the temporal history of individual fs pulses. Drawing on prior work exploring coherent control of atomic wavepackets and chemical processes [122, 123], pulse-shaping with deformable mirrors was combined with an evolutionary search algorithm to temporally shape femtosecond optical pulses in such a way as to preferentially enhance the generation of a particular harmonic order. Experiments with the $q = 23$–33 harmonics of 800 nm in Ar produced by 18 fs pulses resulted in an increase in the intensity of a designated harmonic by up to an order of magnitude, a factor of two to six improvement in the integrated X-ray flux, and spectral widths of individual harmonics approaching the time-bandwidth limit. Similar studies in Kr yielded enhancements of 3–4 in the peak intensity of any of the harmonics between $q = 17$ and 23, whereas the intensity of

adjacent harmonics ($q \pm 2$) was increased by factors of only 1.5–2. It must be emphasized that Bartels et al. [121] also employed a hollow fiber (175 μm i.d.) in the experimental system and, therefore, the reported harmonic intensity enhancements are beyond those available with the fiber alone. Earlier this year, Yoshitomi and co-workers [124] realized enhancements as large as 13 in the single pulse energies of the phase 27th–45th harmonics of 745 nm, generated in Ne by combining adaptive wavefront control with a genetic algorithm. Coherence lengths > 2 mm were observed and the wavefronts of the $q = 39$ and 45 harmonics were flattened significantly by the implementation of adaptive control.

Before leaving this section, other aspects of HHG experiments should be mentioned briefly. Since the motivation for pursuing HHG is, for many applications, producing coherent radiation in the XUV and soft X-ray regions, the base vacuum in the HHG chamber is maintained below 10^{-5} Torr and the optics associated with manipulating and characterizing the harmonic radiation are designed and optimized for the spectral region of interest. A recent example is the introduction by Takahashi et al. [125] of a high damage threshold beamsplitter capable of separating the HHG radiation from the co-propagating pump beam. Consisting of Si or SiC plates (having a surface roughness < 1 nm) and set at Brewster's angle, the beamsplitter withstands peak intensities of 0.8 TW cm^{-2}, and has a reflectivity of 56% (for Si) at 29.6 nm ($q = 27$).

The broad range in photon energy (typically spanning more than two orders of magnitude) over which harmonics are produced also presents a formidable challenge to designing detection systems, particularly if one wishes to accurately record several characteristics of the individual harmonics and the overall spectrum simultaneously. Poletto and colleagues [126, 127] have reported the performance of a grazing-incidence spectrometer/two-dimensional (microchannel) detector combination capable of recording simultaneously the overall spectral profile, angular divergence for individual harmonics, and energy conversion efficiency for harmonic radiation, and to do so on a single shot basis. Details of the design and operation of this system can be found in Refs. [126, 127]. The key point to be made is that the drive for ever-increasing HHG efficiencies at XUV and soft X-ray wavelengths is simultaneously fueling the development of new optical components and systems.

3.1.6. Summary

More than a decade of intensive study of the properties of harmonic radiation produced by $\sim$5–150 fs pulses at $\lambda \cong 800$ nm (Ti:Al_2O_3) have repeatedly demonstrated it to be a high quality coherent source, providing photons in the deep-UV to soft X-ray regions. Energy conversion efficiencies for individual plateau harmonics of $> 10^{-7}$ with a gas jet and sub-50 fs pulses at the fundamental can be improved by at least an order of magnitude with gas-filled hollow fibers, and combined with adaptive optics-based, active control of the laser pulse temporal profile, promises to routinely offer efficiencies exceeding 10^{-5} and single pulse energies of 10 nJ/harmonic (in the cutoff region). Plateau harmonics produced by $\lesssim$25 fs laser pulses exhibit beam divergences of several mrad (less with a hollow fiber) and XUV continua extending to the edge of the water window have been generated with a

beam divergence approaching the diffraction limit. Although a brightness for individual harmonics in the 10–20 nm region exceeding $6 \times 10^{13}\,\mathrm{W\,sr^{-1}\,cm^{-2}}$ for 7 fs pulses with a peak intensity approaching $10^{15}\,\mathrm{W\,cm^{-2}}$ has already been achieved, increasing this value will undoubtedly result from HHG in new fiber designs and the widespread implementation of active control of the laser pulse waveform.

In addition, pulses in the attosecond time domain can be produced with fundamental optical field durations <10 fs and the challenge of the next several years will be to devise techniques by which such pulses can be more thoroughly characterized and their manipulation simplified. On the basis of these results, one might reasonably conclude that HHG offers to the optical physicist or engineer a coherent source in the $\sim$3–300 nm spectral region of unprecedented flexibility in a compact (table-top) package.

3.2. Above threshold ionization (ATI), laser-assisted Auger decay (LAAD), and the laser-assisted photoelectric effect (LAPE)

When a laser pulse having a peak intensity $>10^{12}$–$10^{13}\,\mathrm{W\,cm^{-2}}$ interacts with a gas or solid target, harmonic generation is but one of several phenomena that have been observed. Although HHG is, from the standpoint of its potential applications to materials science, biology, and photochemistry, of primary interest, other high optical field-matter effects to be discussed here often occur concurrently with harmonic generation and are indispensable to gaining a deeper understanding of the physical mechanisms responsible for these and other phenomena. Specifically, ATI, LAAD and the LAPE appear to share a common origin with HHG and, indeed, we might expect to find in them clues to effective routes to probing and optimizing HHG.

The fundamental interaction that is the theme of this section is that between an intense optical field and an electron. Weingartshofer et al. [128] were the first to report the absorption (and emission) of multiple photons by an electron. In the presence of a $\lambda = 10.6\,\mu\mathrm{m}$ (CO_2 laser) optical field having a maximum intensity $>10^{9}\,\mathrm{W\,cm^{-2}}$, electron-Ar atom scattering experiments revealed the existence of "sidebands" in the electron energy spectrum, local maxima separated from the central (zero optical field) peak by multiples of the photon energy. It was noted in Ref. [128] that this optical field-electron interaction, illustrating the exchange of energy between the field and the electron, is clearly a quantum effect that cannot be accurately described by a perturbative theoretical approach.

Since the pioneering experiments of Ref. [128], several related phenomena, all of which are rooted in the interaction between an electron and an intense optical field, have been observed. The foremost of these is ATI, a multiphoton ionization process in which an atom absorbs more photons from the optical field than the minimum necessary to exceed the field-free ionization potential. First observed by Agostini et al. [129] in 1979, ATI is characterized by photoelectron energy spectra comprising a series of peaks separated by integral multiples of the photon energy. Fig. 16, for example, illustrates the photoelectron energy spectrum for Xe when irradiated by 100 ps laser pulses having a peak (focal) intensity of $2 \times 10^{13}\,\mathrm{W\,cm^{-2}}$ and produced by an Nd:YAG laser system ($\lambda = 1.064\,\mu\mathrm{m}$) [130, 131]. Rather than the single feature

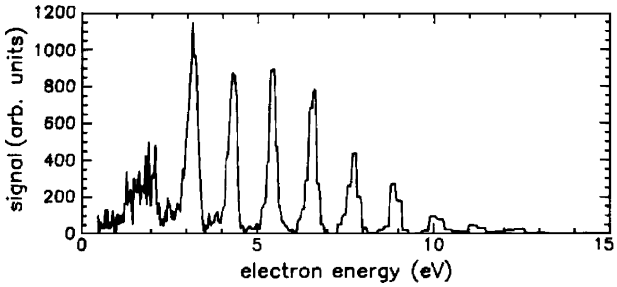


Fig. 16. Photoelectron energy spectrum recorded for Xe photoionized with 100 ps pulses having a wavelength and peak intensity of 1.064 μm (Nd:YAG) and 2×10^{13} W cm^{-2}, respectively (reprinted from Ref. [130] by permission).

near threshold that one would expect for atomic photoionization at lower intensities (i.e., $\lesssim$GWcm^{-2}), the spectrum shows clearly that the photoelectron energies take on discrete values up to and beyond 10 eV. As shown in Fig. 17, similar experiments [132] with an Nd:YLF laser system ($\lambda = 1.05$ μm) operating at a PRF of 1 kHz and an electron spectrometer with a dynamic range of several orders of magnitude confirm that the quantized nature of the photoelectron energy spectrum extends to $\geqslant$25 eV. Data in Fig. 17 are shown for peak laser intensities of 3×10^{13} and 1.9×10^{13} W cm^{-2}. Subsequent studies by Paulus and co-workers [133, 134] examined the behavior of ATI in the rare gases excited by $\lesssim$50 fs, $\sim$15 μJ pulses ($\lambda \sim 630$ or 800 nm) and, once again, discrete structure was observed in the photoelectron energy spectrum for electron energies up to and beyond 50 eV. A detailed description of ATI, including its response to an elliptically polarized optical field, is beyond the scope of this review and, for further details, the reader is encouraged to consult Refs. [129–136]. Nevertheless, it is apparent from the representative energy spectra of Figs. 16 and 17 that ATI and HHG are similar in at least one important respect: both exhibit quantum behavior in that their spectra consist of well-defined features spaced by multiples of the fundamental photon energy. For ATI, this spectral signature is a direct reflection of the interaction of the field with an electron (i.e., the absorption or emission of $n\hbar\omega$, where n is an integer) in the vicinity of an atom.

Two related phenomena that must be mentioned are LAAD and the LAPE. Both underscore the linkage between an intense optical field and an electron through the absorption or emission of one or more photons by the electron in the presence of an atom. In 1994, Schins et al. [137] reported pump-probe experiments in which the $L_{23}M_{23}M_{23}$ Auger process in Ar was studied in the presence of an intense optical field ($\lambda = 800$ nm) produced by a 150 fs Ti:Al_2O_3 laser pulse. Having nominal energies of 205 eV, the Auger electrons resulted from photoionization of Ar with X-rays, generated by a Ga plasma which, in turn, was produced by the pump pulse, and the left-hand portion of Fig. 18 displays the photoelectron spectra recorded for several values of the background (probe) optical field intensity. When the probe pulse is blocked (top spectrum, Fig. 18(a)), the four primary Auger peaks expected are

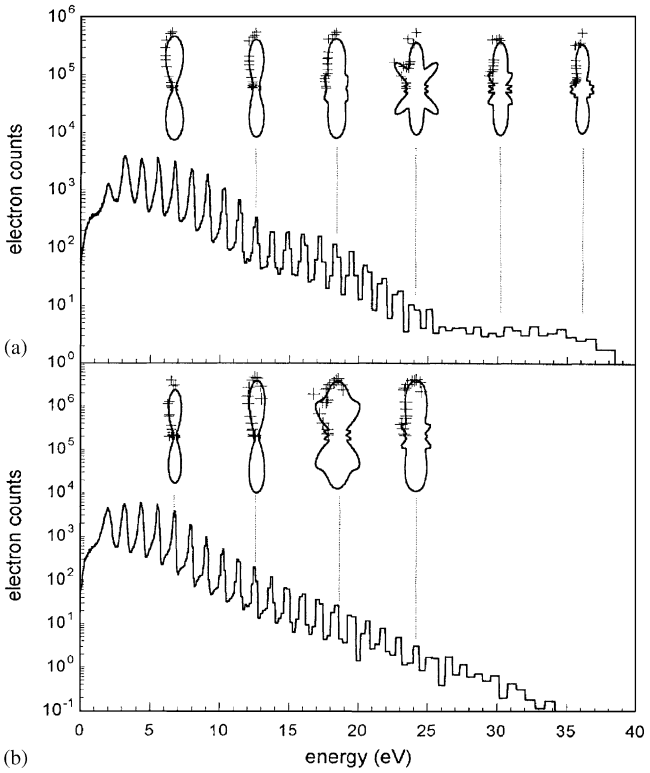


Fig. 17. Xe photoelectron energy spectra acquired with 50 ps laser pulses at a wavelength of 1.05 μm and peak laser intensities of: (a) $3 \times 10^{13}\,\mathrm{W\,cm^{-2}}$ and (b) $1.9 \times 10^{13}\,\mathrm{W\,cm^{-2}}$. The solid curves in the upper portions of both panels illustrate the predicted photoelectron angular distributions for specific ATI peaks (orders) whereas the crosses represent experimental data (reprinted by permission from Ref. [132]).

observed. As the probe field intensity is increased, however, photoelectron sidebands are observed. Fig. 18(b) presents the photoelectron spectra associated with a single Auger line as the near-IR background intensity is increased. The abscissa of Fig. 18(b) is expressed in terms of the number of fundamental (800 nm) photons by which a sideband is removed from the central (zero field) line. As noted by Kienberger et al. [138] recently, "The side bands are a manifestation of the absorption and emission of a discrete number of photons by the electron."

A similar phenomenon was reported almost simultaneously in 1996 by Schins et al. [58] and Glover and co-workers [59] who also observed sidebands in the photoelectron spectra associated with X-ray photoionization of the rare gases. In the experiments of Ref. [59], He was photoionized by soft X-ray ($\hbar\omega \sim 38\,\mathrm{eV}$) pulses provided by HHG in Ar. In the presence of a strong ($\sim 10^{15}\,\mathrm{W\,cm^{-2}}$) optical field having a fundamental wavelength of 800 nm, the photoelectron peaks comprising the zero external field spectrum were weakened significantly and at least seven new photoelectron features, all of which were separated by integral multiples of 1.55 eV

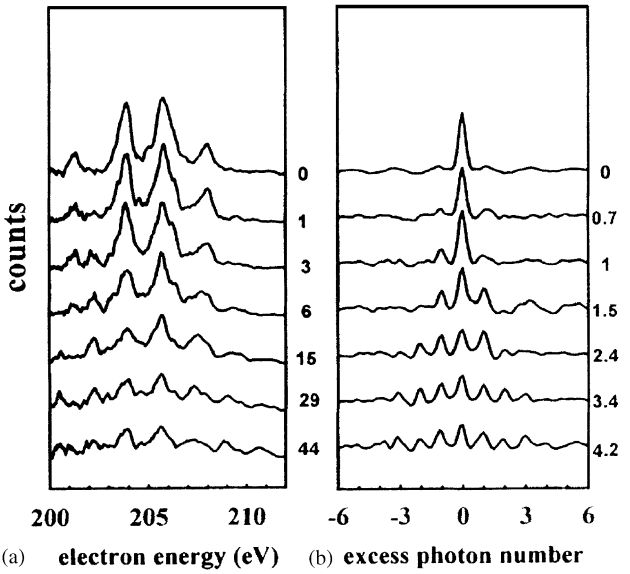


Fig. 18. Photoelectron spectra exhibiting laser-assisted Auger decay (LAAD). In the left-hand portion are shown electron energy spectra for the $L_{23}M_{23}M_{23}$ Auger transition in Ar in the presence of an intense, 150 fs optical field (800 nm). The traces shown are for various values of the near-IR field intensity, in units of 10^{10} W cm^{-2}. The right-hand portion of the figure shows the evolution of the photoelectron spectrum for a single Auger feature as the 800 nm optical field intensity is increased. The right ordinate indicates the number of electron energy sidebands predicted by semiclassical theory for a given optical field intensity (reprinted by permission from Ref. [137]).

(i.e., hc/λ for $\lambda = 800$ nm), appeared. The results of Schins et al. [58] were similar in that He was again photoionized by the $q = 17$–25 harmonics of 800 nm in Ar. Thus, five photoelectron peaks — the signature for the single photon ionization of He by the five harmonics impinging on the He gas jet — were observed in the absence of an external optical field. However, introducing 150 fs, 800 nm pulses into the photoionization region resulted in, for each primary photoelectron peak, the observation of four electron energy sidebands, arising from the absorption or emission of one or more fundamental photons by the photoelectron.

On the basis of the data of Refs. [58, 59, 137], one concludes that the appearance of sidebands in the photoelectron spectra associated with various processes in the presence of an intense optical field, can only result from a direct interaction between the photoelectron and the field. Specifically, the upper and lower sidebands arise from the absorption of one or more photons from the field, or the release of photon(s) back to the field, respectively. Both processes are of interest as we consider the origin of HHG because they are a reminder that, despite the total number of photons in the optical field, the electron–field interaction is inherently quantized and the photon is the "coin of the realm." Thus, although the classical acceleration of an electron in an optical field is clearly valid, it is equally true that the exchange of

energy between the field and electron occurs in integral multiples of the fundamental photon energy. Lastly, it should be emphasized that the phenomena of Refs. [58, 59, 137] occur under conditions in which harmonic generation is observed. Indeed, measurement of the cross-correlation of a single XUV harmonic (or a group of harmonics) with the fundamental, in which the observable is an electron energy sideband [58–60, 137], is now a standard experimental technique for determining the temporal width of harmonic radiation pulses [63, 93, 99]. Under conditions identical to those commonly employed for HHG, must be viewed as a group that provides insight into the physical mechanisms responsible for HHG. It is evident, therefore, that the exchange of a discrete number of photons between an electron and the fundamental optical field is concurrent with HHG. This conclusion provides insight into the origin of HHG which is the subject of the next section.

4. Theory

One of the most remarkable features of HHG is, in addition to its utility as a bright source of high energy (10–500 eV) photons, the fascinating window that it offers onto intense optical field–atom interactions. This section briefly reviews two contrasting views of the physical processes responsible for harmonic generation. The semiclassical view, which has come to be known as the Two Step Process or the Simpleman Model, is intuitively appealing and amenable to calculations. Although remarkably successful in describing a number of the characteristics of HHG, semiclassical models suffer from several drawbacks, foremost of which is the inability to provide a physical mechanism responsible for the clearly quantized nature of both ATI and HHG. A recently developed quantized-optical field approach, which attributes HHG to optical transitions between stationary states of the electron–photon field system appears to meet the necessary requirements and is described briefly in Section 4.2.

Before discussing the semiclassical and quantum pictures of HHG, it should be noted that actually two distinct semiclassical views of HHG have arisen from the research communities pursuing gas and solid conversion media. Although both have advanced a semiclassical description of the phenomenon, the two schools of thought differ markedly concerning the fundamental mechanism responsible for HHG. Emphasis here will be on theoretical efforts devoted to HHG in the gas phase. However, a clear need exists for a unified description of HHG by gas and solid targets since it is quite likely that the same fundamental processes are responsible for HHG, irrespective of the interaction medium.

4.1. Semiclassical picture: two step process

HHG experimental results have, to date, almost universally been interpreted within a classical or semiclassical framework. The most prevalent picture is that of the Two Step (or Simpleman) model [139, 140] in which a bound electron tunnels into the continuum under the influence of the intense, external field. In this

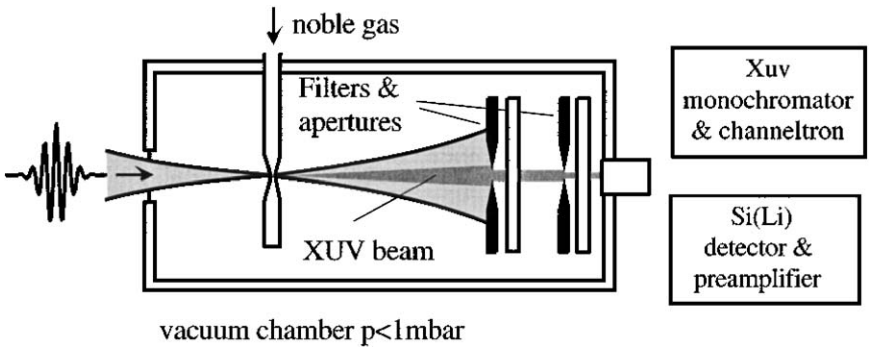


Fig. 19. Sequence of diagrams (reprinted from Ref. [9] by permission) qualitatively portraying the ionization of an atom exposed to increasingly intense optical fields. In all three portions of the figure, the dashed curves represent the Coulombic potential (for zero external field) and $-W_b$ is the atomic ionization potential. For lower field intensities (a), the atom is ionized directly by multiphoton ionization; **(b)** increasing the intensity of the optical field modifies the potential (for the electric field directed to the left) such that the electron is now able to tunnel through the Coulombic barrier; and **(c)** sufficiently high optical field intensities suppress the barrier to the degree that the atom is ionized in the absence of tunneling since the electron energy exceeds the suppressed barrier height.

framework, the optical field serves only to suppress the Coulombic barrier and accelerate (or decelerate) the electron. The interaction of the electron with the optical field is treated classically and radiation is produced when the free electron recombines with the parent ion.

Fig. 19 qualitatively illustrates the dominant processes by which the atom is photoionized as the optical field intensity is increased [9]. In the absence of an optical field, the electron is bound by the Coulombic potential (denoted by the dashed curves in Figs. 19(a)–(c)) and, at comparatively low field intensities, the atom is photoionized by multiphoton ionization, of which ATI is a special case. Continuing to increase the optical field intensity has the effect of modifying, as shown in Fig. 19(b) for the half-cycle of the optical field in which the electric field $\vec{E}$ is directed to the left, the Coulombic potential such that the bound electron is able to tunnel through the potential barrier. At still higher intensities, the Coulombic barrier is suppressed to the extent that the ground state energy of the electron is momentarily above the barrier and the atom is photoionized without the necessity of tunneling (Fig. 19(c)).

Once free of the parent atom, the photoelectron is initially accelerated away from the atom. Reversal of the field direction in the subsequent half-cycle of the optical field re-directs the electron back towards the parent ion, permitting recombination to occur. According to the two-step model view, the electron approaches the ion along one of two primary trajectories and shaping the electron trajectory is tantamount to controlling the HHG process [141]. Several experimental observations, including the temporal coherence of the harmonic radiation [61], structure in ATI photoelectron energy spectra [133], and the HHG process itself [142] have been specifically attributed to these two primary trajectories or to interference between them.

The unifying theme in semiclassical theoretical treatments of HHG is to solve for the dipole moment associated with an electron oscillating in the electric field produced by the fundamental optical field. The formalism briefly reviewed here is that given by Salières et al. [72] which was based on the strong field approximation (SFA) description of the single atom response developed originally by Lewenstein et al. [143] and elaborated in a series of subsequent publications [96, 141, 144]. Once the time-dependent Schrödinger's equation has been solved in the single active electron approximation for the wave function $|\Psi(t)\rangle$ of an atom in an intense optical field (e.g., strong electric field), the dipole moment $\vec{x}(t) = \langle\Psi(t)|\vec{x}|\Psi(t)\rangle$ can be determined. The result is an expression of the form

$$\vec{x}(t) = \mathrm{i}\int_0^\infty d\tau\left(\frac{\pi}{\nu + \mathrm{i}(\tau/2)}\right)^{3/2}\vec{d}^*\left(\vec{p}_s - \vec{A}(t)\right)\exp(-\mathrm{i}S(\vec{p}_s, t, \tau)) \times \vec{E}(t-\tau)\cdot\vec{d}\left(\vec{p}_s - \vec{A}(t-\tau)\right)\exp(-\Gamma t) + \text{c.c.}, \quad (1)$$

where $\vec{A}(t)$ is the vector potential of the fundamental field, $\vec{E}(t) = -\partial\vec{A}/\partial t$ is the electric field, Γ is the ground state ionization rate, ν is a constant, $\vec{p}_s = \int_{t-\tau}^{t} dt''\vec{A}(t'')\tau$ is the value of the electron momentum at the saddle point, and $S(\vec{p}, t, \tau)$ is the quasi-classical action, given by

$$S(\vec{p}, t, \tau) = \int_{t-\tau}^{t} dt''\left(\frac{\left(\vec{p} - \vec{A}(t'')\right)^2}{2} + I_\mathrm{p}\right), \quad (2)$$

in which $\vec{p}$ is the canonical momentum, and $\mathrm{I_p}$ is the atomic ionization potential. As discussed in Ref. [72] (and other work cited above), the integral in Eq. (1) extends over the range in values of time in which the photoelectron is free (i.e., the span in time between photoionization and recombination), and it was found that the shortest electron trajectories are the dominant contributors to the integral of Eq. (1). Only two trajectories have return times less than one period of the optical field.

In this semiclassical theoretical framework, the dipole moment can also be expressed as the linear combination [72]

$$\vec{x}(t) = \sum_{q\ odd}\vec{x}_q\mathrm{e}^{-\mathrm{i}q\omega t - \Gamma t} + \text{c.c.}, \quad (3)$$

where q again denotes the harmonic order, and the coefficients x_q can be computed directly from the Fourier transform of Eq. (1). It is important to recall, however, that the derivation of Eq. (1) assumed the single active electron approximation and, therefore, a complete description of the dipole moment for a given atom requires that one account for all of the active electrons.

Regardless of the physical mechanism(s) responsible for the single atom response, the macroscopic parameters of harmonic radiation, such as the spectral profile and beam quality, are determined by the propagation of the harmonic wave through the active medium. Consequently, it is necessary to solve Maxwell's equations for both the fundamental and harmonics which, if the paraxial and slowly varying envelope

approximations are assumed to be valid, can be written

$$\nabla_{\perp}^{2}\vec{E}_1(\vec{r},t)+2ik_1^0\frac{\partial\vec{E}_1(\vec{r},t)}{\partial z}+2k_1^0\delta k_1(\vec{r},t)\vec{E}_1(\vec{r},t)=0 \tag{4}$$

and

$$\nabla_{\perp}^{2}\vec{E}_q(\vec{r},t)+2ik_q^0\frac{\partial\vec{E}_q(\vec{r},t)}{\partial z}+2k_q^0\left[\Delta k_q^0(\vec{r},t)+\delta k_q(\vec{r},t)\right]\vec{E}_q(\vec{r},t)=-\frac{q^2\omega^2}{\varepsilon_0 c^2}\vec{P}_q^{\mathrm{NL}}(\vec{r},t), \tag{5}$$

where $\vec{E}_1$ and $\vec{E}_q$ represent the envelopes of the fundamental and harmonic electric fields, respectively, $k_q^0=q\omega/c$, and $\delta k_q(\vec{r},t)=-e^2n_e(2m_0qc\omega)^{-1}$ where m_0 is the electron rest mass and n_e is the spatially and temporally dependent electron density. Because this aspect of the theory is not germane to the fundamental physical mechanism(s) by which harmonic radiation is generated, the subtleties will not be described further. For more details concerning the propagation relations, their implications and solutions, the reader is referred to the excellent review by Salières and co-workers [72] and to the references cited therein.

4.2. Optical transitions between stationary states of the photon field-electron system: quantized optical field view of HHG

4.2.1. General Comments

Despite the remarkable success of the semiclassical model of HHG with regard to its ability to qualitatively reproduce experimental spectra, polarization characteristics [74, 145], and harmonic spectral widths [73], for example, several considerations strongly suggest that the semiclassical perspective does not capture key aspects of the physics of the HHG process and one is, therefore, compelled to investigate other theoretical avenues. The foremost of these considerations is that the semiclassical viewpoint is unable to reflect the inherently quantum nature of processes such as ATI and HHG. Since the semiclassical models rely on the acceleration of a free electron in the optical field and the subsequent recombination of that electron with an ion as the radiation production mechanism, they do not offer a physical process for producing the discrete photon and photoelectron energies observed experimentally. As discussed in the last section, virtually all theoretical treatments of HHG have calculated the dipole moment for the electron oscillating in a classical optical field and have associated the terms of the Fourier transform of the dipole moment with the odd harmonics. This approach leaves unresolved the mechanism(s) by which the harmonics are generated and does not account for the quantized nonzero energies of the photoelectrons produced by ATI. Experiments have clearly demonstrated that the energies of photoelectrons produced under the same conditions in which high harmonic radiation is generated are not continuous but rather take on discrete values.

The second drawback of the semiclassical picture is its silence on the subject of the direct, quantized interaction between the electron and the optical field. Certainly, at the heart of the semiclassical models is the force exerted on the electron by the

macroscopic $\vec{E}$ field associated with the optical field. However, the experiments discussed in Section 3.2 confirm that the optical field and photoelectrons interact at the quantum level through the absorption or emission of a discrete number of fundamental photons by the electron. Such an interchange has a profound bearing on the photoelectron energy distribution, but also provides a mechanism for producing photons having energies that are an integral multiple of the fundamental photon energy.

Lastly, a subject that has received scant attention in the discussions concerning HHG is the mechanism by which radiation is emitted in the semiclassical framework — namely, the recombination of a hot electron with the parent ion. Despite the centrality of this process to the semiclassical model, the available experimental evidence appears to contradict the semiclassical picture. The recombination of an electron with an ion is a subject of great interest to the atomic and molecular physics community and has been studied extensively for more than 50 years. Both experiments and theory have demonstrated that recombination is a sensitive function of the electron temperature, and the recombination of a hot electron, in particular, with an ion is a low probability event. Two well-known examples will serve to illustrate this conclusion. Dissociative recombination of molecular ions such as the rare gas dimer ions ($Xe_2^+, Kr_2^+, \ldots$) was investigated in detail for more than two decades because of the role this process plays in mediating the electron density in rare gas plasmas. The point germane to this discussion is that several experiments [146] explored the product states and electron temperature dependence of this mechanism by heating the electrons by inverse bremsstrahlung with microwaves ($\hbar\omega \sim 10^{-4}$–10^{-5} eV) which had the effect of strongly suppressing e^--dimer ion recombination, owing to the $T_e^{-3/2}$ dependence of the recombination rate constant on electron temperature, T_e. Similar comments could be made concerning the recombination process that is the closest analog to that envisioned in the two-step model. Radiative recombination is expressed as

$$A^+ + e^- \rightarrow A^* + \hbar\omega, \tag{6}$$

where the asterisk denotes an excited state of the atom A. As discussed in detail by Biondi [147], this process is "…rather unlikely…" because "…the probability of radiative capture depends on the likelihood of a free-bound radiative transition occurring during the electron-ion encounter (collision). A thermal electron will negotiate the region around the ion…in a time $t_{coll} \sim R/v = 10^{-14}$ s" [147]. The capture probability for electrons of this energy is $\sim 10^{-6}$–10^{-5} which corresponds to a cross-section of 10^{-19} cm^2. Quantum calculations indicate that "…for electron capture into the lower levels of the excited atom the partial recombination coefficient varies as $T_e^{-0.5}$…" Thus, the recombination coefficient for a 50 eV electron can be expected to be a factor of ~40–45 smaller than that for a thermal electron. One must hasten to add that the presence of the intense optical field may alter this situation by driving the recombination process and further theoretical and experimental scrutiny is necessary to determine if this process plays a significant role in HHG.

One other experimental observation that bears mentioning is the presence of recombination emission *following* harmonic generation. Crane et al. [36], for

example, clearly observe the 1s–*n*p series of hydrogen-like He^+ when harmonics (up to $q=45$) of 526 nm are produced at intensities $>10^{17}\,W\,cm^{-2}$. Streak camera data show recombination emission produced for several ns after the optical driver pulse has terminated. The important point to be made is that HHG and recombination emission are produced under identical experimental conditions but differ fundamentally in their temporal behavior. Harmonic generation occurs only during the fundamental optical pulse whereas recombination-produced emission is substantially time-delayed. These results are consistent with experiments of the last 30 years in which the recombination of singly- or multiply-charged ions has served as the pumping mechanism for short wavelength (XUV, soft X-ray) lasers [148, 149].

This background *appears to contradict* the semiclassical model which envisions the recombination of electrons having energies of tens to hundreds of eV with an ion, thereby converting kinetic energy into a VUV or XUV photon. None of the arguments presented here are intended to suggest that the physical picture presented by semiclassical models does not valid but they do imply that: (1) one or more other processes are operative in HHG and, perhaps, dominant; (2) the subject of the underpinnings of HHG warrants further theoretical and experimental attention.

4.2.2. Quantum Volkov state theory of harmonic generation

An alternative physical picture of HHG that accounts for all of the observations cited above was introduced by Gao et al. [150] in 1998 and described in further detail in Refs. [86, 145, 151]. In this picture, the optical field assumes a more active role in the process than is suggested by the semiclassical theory. Radiation is produced by transitions between quantized field (quantum) Volkov states which are *stationary* solutions of Schrödinger's equation for a quantized optical field interacting with an electron. Once injected into the optical field by ATI, photoelectrons interact with photons to form Volkov states and the HHG spectrum, for example, can be expressed in terms of the Volkov-to-Volkov state transition probability. Since quantum Volkov states are stationary, the total energy and momentum of the electron–photon system are well-defined and such Volkov states may be regarded as the high optical field analog of bound states of an atom. The electron–photon interaction is dominant, which is to be expected for the optical intensities characteristic of HHG experiments and is inevitable as peak intensities of the driving optical field continue to rise. Energy and momentum conservation, the generation of odd harmonics (rather than broadband radiation), and the spectral and polarization characteristics of each harmonic follow directly from the theory, which has no adjustable constants.

This treatment presents a picture of the mechanisms responsible for HHG that contrasts with the semiclassical view. Specifically, the discrete photoelectron energies produced by ATI are preserved throughout the HHG process; that is, the electron maintains its momentum during Volkov-to-Volkov state transitions. Also, HHG is an optical energy-conversion process in which q pump (fundamental) photons are annihilated to produce a single, harmonic ($q\hbar\omega$) photon. In this sense, it is related to nonlinear Compton scattering by which the conversion of as many as four fundamental photons into a single γ-ray photon has been observed [152]. The

primary conclusion is that, although energy and momentum conservation require that the electron ultimately return to the ground state of the parent atom, the generation of harmonic radiation (including those harmonics having photon energies $q\hbar\omega$ comparable to or less than the atomic ionization potential) results from transitions between quantum Volkov states.

The following paragraphs will review a few of the highlights in the derivation of the matrix element for Volkov→Volkov state transitions which, in turn, yields an expression for the generation rate for the qth harmonic. Much of the discussion will closely follow the presentation of Ref. [151].

If the Coulombic interaction between an electron and its parent ion are ignored, the matrix element for *above threshold ionization* can be expressed as [153]

$$T_{\mathrm{fi}} = \sum_{\vec{P}_1,n} \langle \Phi_{\mathrm{f}}; n|\Psi_{\boldsymbol{\mu}}\rangle \frac{\mathrm{i}\alpha}{E - E_{\boldsymbol{\mu}} + \mathrm{i}\alpha} \langle \Psi_{\boldsymbol{\mu}}| V_1 |\Phi_{\mathrm{i}}; \ell\rangle, \tag{7}$$

where $\vec{P}_1$ is the momentum of an ATI photoelectron, Φ_{i} is a bound atomic state that is decoupled from the photon state $|\ell\rangle$ of the fundamental optical field, Φ_{f} is the final state that is also decoupled from the fundamental field, α is the spectral linewidth of the fundamental radiation, and the electron–optical field interaction is given by

$$V_1 = -\frac{e}{m}\vec{A}(\vec{k}\cdot\vec{r})\cdot(-\mathrm{i}\vec{\nabla}) + \frac{e^2A^2(\vec{k}\cdot\vec{r})}{2m}, \tag{8}$$

in which the vector potential is related to the phase of the photon field, Θ, by the expression

$$\vec{A}(\vec{k}\cdot\vec{r}) = g\left(\hat{\varepsilon}\mathrm{e}^{\mathrm{i}\vec{k}\cdot\vec{r}}\mathrm{e}^{\mathrm{i}\Theta}\hat{a} + \hat{\varepsilon}^*\mathrm{e}^{-\mathrm{i}\vec{k}\cdot\vec{r}}\mathrm{e}^{-\mathrm{i}\Theta}\hat{a}^\dagger\right). \tag{9}$$

In Eq. (9), g is the normalization factor arising from quantization of the optical field and $\hat{\varepsilon}$ is the polarization vector. The crucial feature of Eq. (7) is the introduction of quantized field (quantum) Volkov states Ψ_{μ} (Ref. [154]) as intermediate states in the overall process where the wavefunctions

$$|\Psi_{\boldsymbol{\mu}}\rangle = \left|\vec{P}_s, n_s\right\rangle = V_{\mathrm{e}}^{-1/2}\mathrm{e}^{\mathrm{i}(\vec{P}_s + z\vec{k})\cdot\vec{r}}\mathrm{e}^{-iN_{\mathrm{a}}\vec{k}\cdot\vec{r}}\mathrm{e}^{-F_s}|n_s\rangle \tag{10}$$

are the solutions of the equation

$$(K + V_1)|\Psi_{\boldsymbol{\mu}}\rangle = E_{\boldsymbol{\mu}}|\Psi_{\boldsymbol{\mu}}\rangle = E_{\boldsymbol{\mu}}\left|\vec{P}_1, n_1; 0\right\rangle, \tag{11}$$

in which $K = (-\mathrm{i}\vec{\nabla})^2/2m + N_{\mathrm{a}}\omega$ is the non-interacting Hamiltonian for the electron and optical field, m is the electron rest mass, N_{a} is the number operator for the photon field, V_{e} is the normalization volume for the electron, z is the ponderomotive energy ($U_{\mathrm{p}} = e^2E^2(4m\omega^2)^{-1}$) normalized to the photon energy, ω is the frequency of the fundamental field, E is again the classical amplitude of the electric field, the subscript "s" in Eq. (10) denotes a specific Volkov state ($s = 1, 2, \ldots$), and the coupling between the optical field and the electron is represented by the operator $\exp(-F_s)$, where

$$F_s = \frac{eg}{m\omega}\left(\vec{P}_s\cdot\hat{\varepsilon}\mathrm{e}^{\mathrm{i}\Theta_s}\hat{\mathrm{a}} - \vec{P}_s\cdot\hat{\varepsilon}\mathrm{e}^{-\mathrm{i}\Theta_s}\hat{a}^\dagger\right) + \frac{e^2g^2\cos\xi}{4m\omega}\left(\mathrm{e}^{-\mathrm{i}2\Theta_s}\hat{a}^{\dagger 2} - \mathrm{e}^{\mathrm{i}2\Theta_s}\hat{a}^2\right). \tag{12}$$

In Eqs. (7)–(12), $\hbar \equiv c \equiv 1$, and ξ expresses the degree of polarization of the optical field. The latter is related to the ellipticity μ by the relation

$$\mu = \tan\frac{\xi}{2}. \tag{13}$$

It is important to remember that the Ψ_μ are the (stationary) solutions of the time-independent Schrödinger's equation (Eq. (11)) and that the eigenenergies of the quantum Volkov states are given by

$$E_{\boldsymbol{\mu}} = \frac{\left|\vec{P}_1\right|^2}{2m} + n\omega + z\omega. \tag{14}$$

Note that the energy eigenvalues depend on both the ponderomotive potential and the ATI electron energy but, more importantly, differ by integral multiples of the fundamental field photon energy. This characteristic of quantum Volkov states is matched by the quantized nature of ATI, LAPE, and LAAD photoelectron energy spectra, for example, in which peaks are separated by multiples of the photon energy. A similar statement is true of HHG radiation spectra. This correlation is a direct reflection of the centrality of the quantum Volkov state to all of these intense-field processes.

Although Eq. (7) provides an accurate description of the ATI process (and photoionization, in general), it offers no means for generating a secondary radiation (harmonic) field and is, therefore, not suitable for representing HHG. Since the harmonic intensities are generally considerably smaller than that for the fundamental, a secondary radiation field can readily be introduced by invoking time-independent perturbation theory. Doing so results in the following expression for the perturbed Volkov state, Ψ'_μ,

$$\Psi'_\mu = -\sum_v \frac{|\Psi_v\rangle\langle\Psi_v|V'|\Psi_\mu\rangle}{E_v - E_{\boldsymbol{\mu}}}, \tag{15}$$

in which $|\Psi_v\rangle$ are the quantum Volkov state solutions for the equation:

$$(K + V_2)|\Psi_v\rangle = E_v|\Psi_v\rangle = E_v\left|\vec{P}_2, n_2; 1\right\rangle \tag{16}$$

and whose eigenenergies are given by

$$E_v = E\left(\left|\vec{P}_2, n_2; 1\right\rangle\right) = \frac{\left|\vec{P}_2\right|^2}{2m} + n_2\omega + \frac{e^2g^2n_2}{m} + \omega'. \tag{17}$$

Note in Eqs. (16) and (17) that one photon is now generated in the secondary field and the harmonic frequency is denoted ω'. Upon substituting $\Psi_\mu + \Psi'_\mu$ for Ψ_μ into Eq. (7), the transition matrix for the dominant term contributing to harmonic generation (denoted $T_{\mathrm{fi}}^{\mathrm{HG}}$) is expressed as [145, 150]

$$T_{\mathrm{fi}}^{\mathrm{HG}} = \sum_{(\vec{P}_1,n_1)(\vec{P}_2,n_2)} \left\langle \Phi_{\mathrm{f}}; m; 1|\vec{P}_2, n_2; 1\right\rangle \times \left\langle \vec{P}_2, n_2; 1\left|V'\right|\vec{P}_1, n_1; 0\right\rangle \left\langle \vec{P}_1, n_1; 0|V_1|\Phi_{\mathrm{i}}; l; 0\right\rangle, \tag{18}$$

where primes denote quantities associated with the secondary radiation field, the subscripts 1 and 2 represent parameters prior to and following the Volkov → Volkov

state transition, respectively, V_2 differs from V_1 only by the shift in the phase of the fundamental field, $\Delta\Theta = \Theta_2 - \Theta_1$, that occurs as a result of the harmonic generation (electron–photon field scattering) process, and the interaction term V' is given by

$$V' = -\frac{e}{m}\vec{A}'(\vec{k}'\cdot\vec{r})\cdot(-\mathrm{i}\vec{\nabla}) + \frac{e^2}{m}\vec{A}(\vec{k}\cdot\vec{r})\cdot\vec{A}'(\vec{k}'\cdot\vec{r}). \tag{19}$$

The second term on the right-hand side of Eq. (19) is of particular importance because it describes the interaction between the fundamental and harmonic radiation fields. Generally, when the electron–field interaction is described by the minimum coupling expression $\left(\vec{P} - e\vec{A} - e\vec{A}'\right)^2/2m = \ldots - (e/m)\vec{P}\cdot\vec{A} - (e/m)\vec{P}\cdot\vec{A}' + (e^2/2m)\vec{A}\cdot\vec{A}'$, the last term is ignored. However, the intensities of both the fundamental and harmonic fields that are now available experimentally argue for retention of the $\vec{A}\cdot\vec{A}'$ term. It is this second-order term that represents the exchange of energy between the fundamental and secondary optical fields by means of quantum Volkov states.

The middle term of Eq. (18), denoted T^{V}, is the matrix element that describes a new mechanism for the generation of harmonic radiation — optical transitions between quantum Volkov states. There is no classical analog to this mechanism and it represents a clear departure from the semiclassical picture of harmonic generation in which harmonic radiation is produced by the recombination of an electron with the parent ion. Conserving momentum and energy in the harmonic generation process requires Φ_{f} in Eq. (18) to be the initial, bound state of the atom (or molecule). This constraint requires that $P^{\rightharpoonup}_2 = \vec{P}_1 \equiv \vec{P}$. Therefore, this theory predicts that the electron momentum $\vec{P}$ (determined by the number of photons absorbed during ATI) is not changed by Volkov → Volkov state transitions. Secondly, it is the quantum Volkov state that is responsible for converting fundamental photons into harmonic photons. To summarize, if the interaction of an electron with an optical field is expressed in terms of quantum Volkov states, then introducing a secondary photon field leads directly to the result that the harmonic (secondary) photons are produced by transitions between the quantum Volkov states. This is the single most significant result of quantizing the fundamental photon field.

Evaluating $T_{\mathrm{fi}}^{\mathrm{HG}}$ requires summing over $\vec{P}$, n_1 and n_2. For a specified harmonic $q = n_1 — n_2$, the harmonic generation transition matrix can be written

$$\begin{aligned}
T_q^{\mathrm{HG}} = {} & \mathrm{e}^{\mathrm{i}q\left(\frac{\Delta\Theta}{2}+\phi\right)} \sum_j \left(\frac{g'\omega^2}{2E}\right)(z-j)\sqrt{2m^3\omega^5(j-z-I_{\mathrm{p}})} \\
& \times \int_0^{2\pi} \mathrm{d}\psi \int_0^{\pi} \sin\theta\, \mathrm{d}\theta \left|\mathfrak{I}_{-j}(\zeta_j,\eta,\phi)\right|^2 \left|\Phi_i\left(\vec{P}_j + z\vec{k} - j\vec{k}\right)\right|^2 \\
& \times \left\{ -\frac{2eE}{m\omega^2}\vec{P}_j\cdot\hat{\varepsilon}'^{*}\mathfrak{I}_{-q}\left(x_j, y, -\phi - \frac{\pi}{4}\right)\right. \\
& + \frac{2e^2E^2}{m\omega^3}\mathrm{e}^{-\mathrm{i}\left(\frac{\Delta\Theta}{2}+\phi\right)}\mathfrak{I}_{-q+1}\left(x_j, y, -\phi - \frac{\pi}{4}\right)\hat{\varepsilon}\cdot\hat{\varepsilon}'^{*} \\
& \left. + \frac{2e^2E^2}{m\omega^3}\mathrm{e}^{\mathrm{i}\left(\frac{\Delta\Theta}{2}+\phi\right)}\mathfrak{I}_{-q-1}\left(x_j, y, -\phi - \frac{\pi}{4}\right)\hat{\varepsilon}^{*}\cdot\hat{\varepsilon}'^{*} \right\},
\end{aligned} \tag{20}$$

where j is an integer representing the number of fundamental photons absorbed in the ATI process, I_p is the ionization potential of the atom or molecule of interest, $\Phi_i(\vec{P}_j + z\vec{k} - j\vec{k})$ is the momentum-space representation (Fourier transform) of the bound atomic state wavefunction, $\Im$ is the generalized Bessel function [145], and

$$x = \left|\frac{4eE}{m\omega^2}\vec{P}_1 \cdot \hat{\varepsilon}\right| \sin\frac{\Delta\Theta}{2}, \tag{21}$$

$$y = -z \cos\xi \sin\Delta\Theta, \tag{22}$$

and

$$\phi = \mathrm{Arg}(\vec{P}_1 \cdot \hat{\varepsilon}). \tag{23}$$

The harmonic generation rate for the qth harmonic is then given by

$$W_q(\Delta\Theta) = \frac{V'_\gamma}{(2\pi)^3}\int \mathrm{d}\Omega_{\vec{k}'} 2\pi \left|T_q^{\mathrm{HG}}\right|^2 = \frac{V'_\gamma}{2\pi^2}\int_0^{2\pi} \mathrm{d}(\Delta\Theta)\left|T_q^{\mathrm{HG}}\right|^2, \tag{24}$$

where V'_γ is the photon normalization volume, and $\vec{k}' = q\vec{k}$ denotes the propagation vector for the harmonic optical field.

With the analytic expressions of Eqs. (20) and (24), one is able to calculate the harmonic spectrum or characteristics (polarization behavior, emission spectral lineshape) of a specific harmonic. Each is expressed in terms of the probability associated with transitions between pairs of quantum Volkov states. Detailed comparisons of experimental measurements of HHG spectra, polarization data, and linewidths of individual harmonics with theoretical predictions (Eqs. (20)–(24)) [86, 145, 150, 151] have shown broad agreement between the two. One example is given in Fig. 20 which

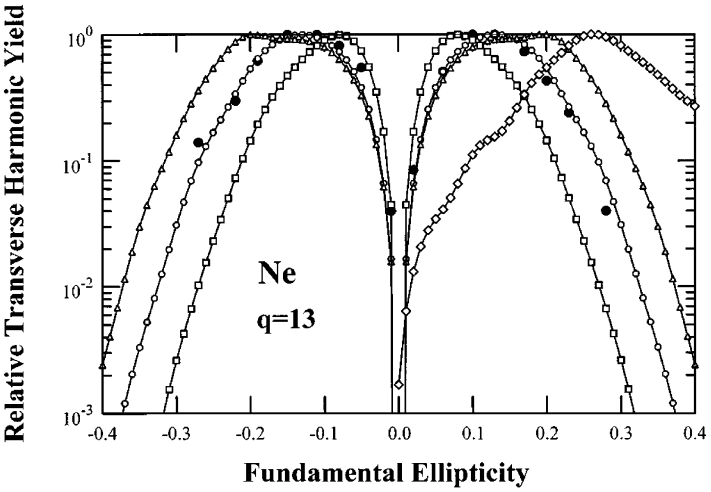


Fig. 20. Comparison of the relative yield of the transverse component of the 13th harmonic of 754 nm in Ne with the experimental data (●) of Ref. [79] (reprinted by permission from Ref. [145]). The fundamental field intensity is 10^{15} W-cm^{-2} and the results of calculations are given for four sets of photoelectron (ATI) orders: $j = 46$(□); $j = 46,47$ (○); $j = 46$–48(△) and $j = 46$–71 (◇).

compares the predictions of Eq. (24) with experimental data of Burnett et al. [79], who characterized the polarization ellipse for specific harmonics as the ellipticity of the driving optical field was varied. The solid circles (•) of Fig. 20 represent measurements (obtained for the $q = 13$ harmonic of 754 nm in Ne and a fundamental intensity of $10^{15}\,W\,cm^{-2}$) of the relative photon yield *transverse* to the major axis of the polarization ellipse of the *fundamental*. The open symbols reflect theoretical predictions when different photoelectron orders (j values) are incorporated into the calculations. Including only two photoelectron orders ($j = 46,47$) yields a close match between theory and experiment, a conclusion consistent with other observations ([86], [145], [150], [151]) suggesting that the lowest energy photoelectrons contribute disproportionately to HHG, particularly for fundamental optical field pulse widths > 100 fs. These results and others cited above suggest that, although differing considerably from the semiclassical view of the HHG process, the attribution of HHG to Volkov-to-Volkov state transitions provides valuable insight into this intriguing phenomenon.

5. Summary and conclusions

Since 1987, a period scarcely longer than 17 years, HHG has progressed from a novel, high intensity nonlinear effect — dramatic if only in view of the extraordinarily high order of the nonlinearity — to a reliable, table-top source of photons having energies extending into the soft X-ray region. Harmonic generation with sub-50 fs, multi-kHz Ti:Al_2O_3 sources now offers beams having a divergence approaching the diffraction limit, and energy conversion efficiencies improving rapidly ($> 10^{-5}$) over the past several years owing to improved phase matching with gas-filled hollow fibers and optimization of the temporal profile of the fundamental optical pulse with adaptive optics control. Although it would not be accurate to suggest that this unique source is inexpensive, the technology of ultrafast laser sources has improved to the point that reliable systems are available commercially, enabling HHG to be applied to fundamental research in photochemistry, materials science, and spectroscopy. Nevertheless, if HHG is to have an impact in the VUV and XUV spectral regions comparable to that of the N_2 laser in the UV (337 nm) in the 1960s and 1970s, the cost of an HHG system must be reduced by a factor of five or more in the years ahead. Further improvements in the efficiency of HHG and the ability to favor the generation of a single harmonic are undoubtedly forthcoming since earlier advances in HHG have galvanized laser researchers and engineers worldwide into pursuing Ti:Al_2O_3 laser and coherent control technology.

It was recognized soon after the initial observations of HHG that the unusually high order of the process renders conventional perturbative theoretical methods untenable and requires a non-perturbative approach to describe not only HHG but associated phenomena as well. HHG experimental results have almost universally been described in terms of the semiclassical two-step, or Simpleman, model in which the electric field is essentially an external agent that serves to suppress the ionization barrier and accelerates an electron. However, the observation of LAAD and LAPE, under the same experimental conditions in which HHG occurs, emphasizes the close

coupling between the photon field and the photoelectron, providing support for a recently advanced, alternative view of HHG in which optical transitions between quantum Volkov states are primarily responsible for the efficient generation of high harmonics. In particular, LAAD and LAPE yield photoelectron spectra that offer definitive evidence of energy exchange between a free electron and an intense optical field. The medium of exchange is $n\hbar\omega$, where n is an integer and $\hbar\omega$ is the photon energy of the fundamental optical field. Of particular interest to HHG are the lower energy sidebands which are the result of an electron releasing one or more photons to the optical field. It appears certain that photoelectrons produced by ATI or tunneling interact with the optical field to produce harmonic photons and, therefore, are at least partially responsible for HHG. An improved understanding of the physical processes undergirding HHG, combined with continuing advances in laser driver technology and gas phase nonlinear conversion schemes, will likely broaden its appeal to researchers in the life and chemical sciences and solidify HHG's position as the preferred laboratory source of coherent radiation in the wavelength region below ~ 150 nm.

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Erratum

Erratum to: "High-order harmonic generation and other intense optical field–matter interactions: Review of recent experimental and theoretical advances [JPQE, 28(1–3)197–246]"

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The following errors occurred in the above paper:

Figs. 14 and 19 were transposed in the original article. The correct figures and captions are produced here in their entirety:

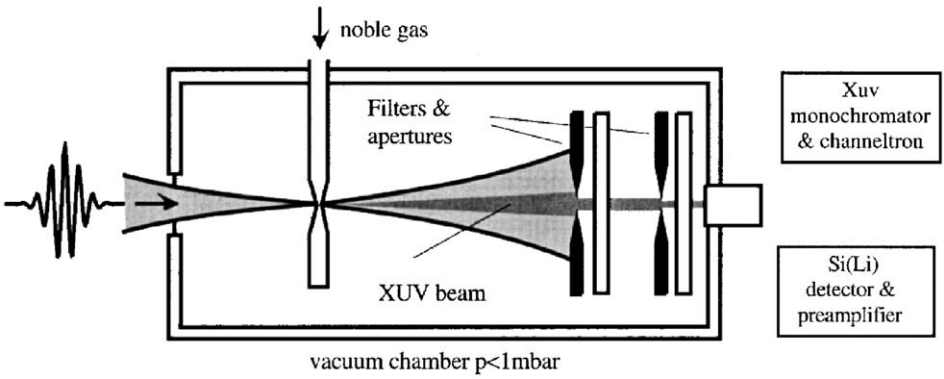


Fig. 14. Schematic diagram of a representative apparatus for generating and detecting harmonic radiation (reprinted from Ref. [9] by permission). In this system, the backing pressure for the rare gas jet is ~2.5 atm whereas the base pressure for the vacuum system is typically $<10^{-5}$ Torr.

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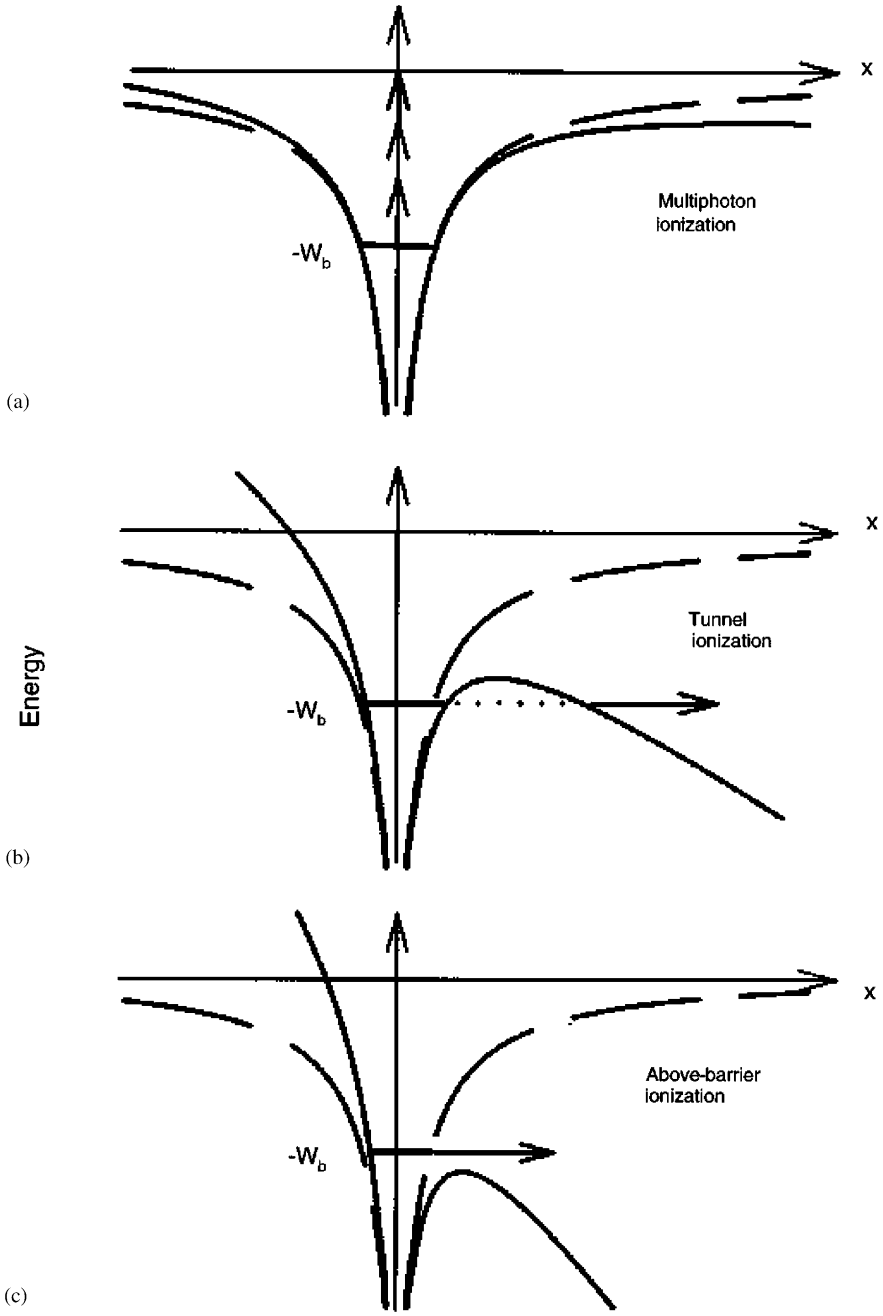


Fig. 19. Sequence of diagrams (reprinted from Ref. [9] by permission) qualitatively portraying the ionization of an atom exposed to increasingly intense optical fields. In all three portions of the figure, the dashed curves represent the Coulombic potential (for zero external field) and $-W_b$ is the atomic ionization potential. For lower field intensities (a), the atom is ionized directly by multiphoton ionization; (b) increasing the intensity of the optical field modifies the potential (for the electric field directed to the left) such that the electron is now able to tunnel through the Coulombic barrier; and (c) sufficiently high optical field intensities suppress the barrier to the degree that the atom is ionized in the absence of tunneling since the electron energy exceeds the suppressed barrier height.

Fig. 6 was printed in B/W and is produced here in color:

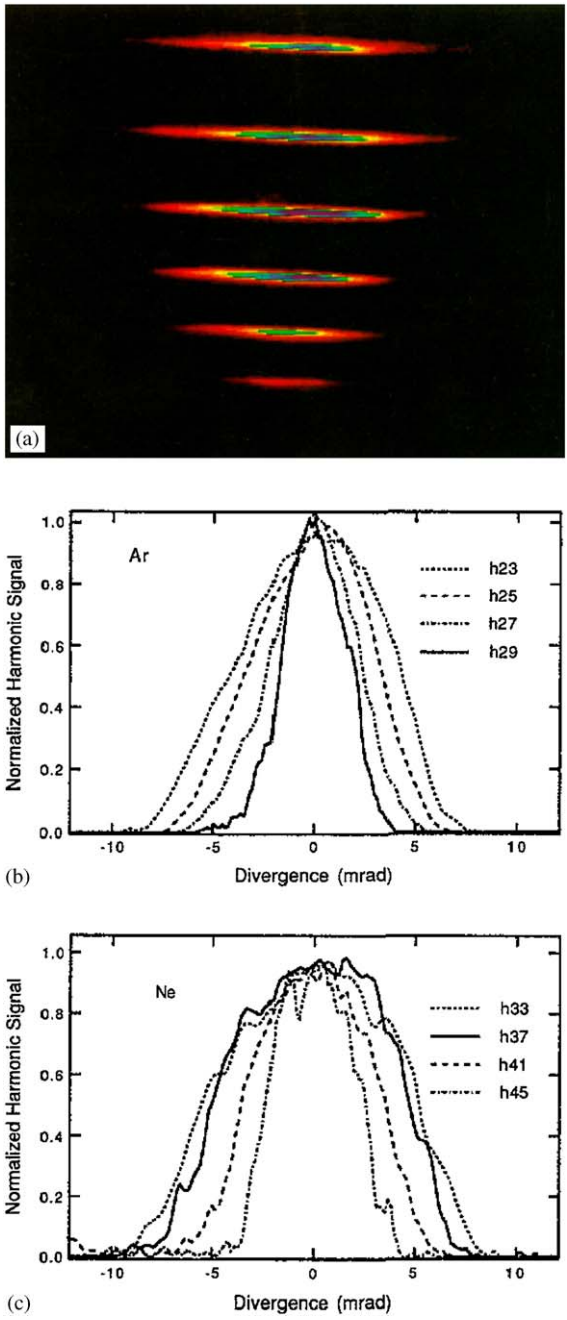


Fig. 6. Cross-sectional (transverse) intensity profiles for several harmonics of the Cr:$LiSrAlF_6$ laser ($\lambda = 825$ nm) in Ar and Ne: (a) image acquired with a spectrometer and CCD camera of the $q = 19$–29 harmonics produced in Ar at an intensity of 1.8×10^{14} W cm^{-2}. The vertical axis of the image indicates wavelength (increasing upwards) whereas the spatial variation of the intensity of a given harmonic is oriented along the horizontal axis of the image; (b) transverse intensity profiles for the 23rd, 25th, 27th and 29th harmonics generated in Ar at a peak fundamental intensity of 1.8×10^{14} W cm^{-2}; (c) $q = 33$, 37, 41, and 45 produced in Ne for a peak intensity of 2.7×10^{14} W cm^{-2} (reprinted from Ref. [67] by permission).

1. p. 208, lines 1 and 2, should read "...the photon yield in the plateau (in Ne) fell, resulting in pulse energies of 6-8 pJ/harmonic and an energy conversion efficiency of $\sim 10^{-10}$."

2. p. 216, line 10, should read "...polarization behavior of HHG provides a window into the nature of the harmonic..."

3. p. 226, line 4-9, should read: "...are beyond those available with the fiber alone. Earlier this year, Yoshitomi and co-workers [124] realized enhancements as large as 13 in the single pulse energies of the 27th-45th harmonics of 745 nm, generated in Ne by combining adaptive wavefront control with a genetic algorithm. Coherence lengths >2 mm were observed and the phase wavefronts of the $q = 39$ and 45 harmonics were flattened significantly by the implementation of adaptive control."

4. p. 231, line 7: The sentence that begins "Under conditions identical..." should read: "A variety of phenomena, observed under conditions identical to those commonly employed for HHG, must be viewed as a group that provides insight into the physical mechanisms responsible for HHG."

5. p. 236, line 15, should read "...by semiclassical models is not valid but they do imply that : (1) one or more other..."

6. p. 236, line 20, should read: "This constraint requires that $\vec{P}_2 = \vec{P}_1 = \vec{P}$. Therefore, ..."